An Introduction to Causal Inference

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Day 1

Part 1: Introduction

LaLonde (1986): (causal) effect of a job training program on earnings (I)

- Randomized evaluation of a large scale job training program NSW
- NSW: National Supported Work Demonstration

```
> #### Data (Lalonde's randomized study)
> data(lalonde, package = "Matching")
> head(lalonde)
age educ black hisp married nodegr re74 re75     re78 u74 u75 treat
1  37   11     1    0       1      1    0    0  9930.05   1   1     1
2  22    9     0    1       0      1    0    0  3595.89   1   1     1
3  30   12     1    0       0      0    0    0 24909.50   1   1     1
4  27   11     1    0       0      1    0    0  7506.15   1   1     1
```

treat: (1:treated / 0:control);

age & educ: measured in years;

black: (1: black, 0: otherwise);

hisp: (1: Hispanic, 0 otherwise);

married: (1: married, 0: otherwise);

nodegr: high school diploma (1: no degree, 0: otherwise);

re74, re75: real earnings in 1974/1975;

re78: real earnings in 1978.

REMARKS:

- NSW $\Rightarrow$ 6–18 months of work experience;
- Target: individuals facing economic and social problems prior to enrollment in the program;
- Eligible candidates were randomized between March-1975 and July-1977;

LaLonde (1986): (causal) effect of a job training program on earnings (II)

Effect Evaluation 1: Difference of Means (From the randomized data)

```
> mean(lalonde$re78[lalonde$treat==1]) - mean(lalonde$re78[lalonde$treat==0])
[1] 1794.343
```

$\Rightarrow$ $\text{ATE} = 1,794.34$, with C.I. (95%): $[474.01;\ 3,114.68]$

Effect Evaluation 2: Linear Regression ("Conditional")

```
> lalonde.fit <- lm(re78 ~ ., data=lalonde)
> round(summary(lalonde.fit)$coef[12,], 3)
Estimate Std. Error    t value   Pr(>|t|)
1670.709    641.132      2.606      0.009 **
```

$\Rightarrow$ $\text{ATE} = 1,670.71$

- What if we use only the covariates `treat` and `age?` ("Partially Conditional")

```
> lalonde.fit2 <- lm(re78 ~ treat + age, data=lalonde)
Estimate Std. Error    t value   Pr(>|t|)
1758.575    633.647      2.775      0.006 **
```

- The estimated effects are quite similar! Why? Should this always be the case?

LaLonde (1986): (causal) effect of a job training program on earnings (III)

Covariate Balancing:

```
> vars <- c("age","educ","black","hisp","married","nodegr","re74","re75","re78","
  u74","u75")
> group_var <- "treat"
> tbl <- tableone::CreateTableOne(vars=vars, strata=group_var, data=lalonde)
> print(tbl)
Stratified by treat
                 0                  1            p-test
n              260                185
age          25.05 (7.06)       25.82 (7.16)      0.265
educ         10.09 (1.61)       10.35 (2.01)      0.135
black         0.83 (0.38)        0.84 (0.36)      0.649
hisp          0.11 (0.31)        0.06 (0.24)      0.076
married       0.15 (0.36)        0.19 (0.39)      0.327
nodegr        0.83 (0.37)        0.71 (0.46)      0.001
re74       2107.03 (5687.91) 2095.57 (4886.62)   0.982 (EARNINGS BEFORE TREATMENT)
re75       1266.91 (3102.98) 1532.06 (3219.25)   0.382 (EARNINGS BEFORE TREATMENT)
re78       4554.80 (5483.84) 6349.15 (7867.40)   0.005 <----- (!!)
u74           0.75 (0.43)        0.71 (0.46)      0.326
u75           0.68 (0.47)        0.60 (0.49)      0.065
```

⇒ Covariates are distributed almost equally across groups.

- This is the essence of causal inference...
  (so what happens when that doesn't happen?)

LaLonde (1986): (causal) effect of a job training program on earnings (IV)

MODIFIED DATASET: inclusion of new units from (https://www.census.gov/)

Current Population Survey (CPS)

The Current Population Survey (CPS), sponsored jointly by the U.S. Census Bureau and the U.S. Bureau of Labor Statistics (BLS), is the primary source of labor force statistics for the population of the United States.

Inclusion rule: CPS males under age 55, not working when surveyed in March 1978, unemployed in 1976 and below poverty level in 1975.

```
> data(lalonde, package = "designmatch")
> dim(lalonde)
[1] 614  10
> names(lalonde)
[1] "treatment" "age" "education" "black" "hispanic" "married" "nodegree" "re74"
 "re75" "re78"
> # Including new variables:
> lalonde$u74 <- as.numeric(lalonde$re74==0)
> lalonde$u75 <- as.numeric(lalonde$re75==0)
> head(lalonde)
  treat age educ black hispanic married nodegree re74 re75       re78 u74 u75
1     1  37   11     1        0       1        1    0    0  9930.0460   1   1
2     1  22    9     0        1       0        1    0    0  3595.8940   1   1
3     1  30   12     1        0       0        0    0    0 24909.4500   1   1
4     1  27   11     1        0       0        1    0    0  7506.1460   1   1
```

LaLonde (1986): (causal) effect of a job training program on earnings (V)

Covariate Balancing:

```
> vars <- c("age","education","black","hispanic","married","nodegree","re74","re75","
    re78")
> group_var <- "treatment"
> table.obj <- tableone::CreateTableOne(vars = vars, strata = group_var, data =
    lalonde)
> print(table.obj)
Stratified by treatment
                 0                  1            p-test
n              429                185
age          28.03 (10.79)      25.82 (7.16)      0.011 *
education    10.24 (2.86)       10.35 (2.01)      0.633
black         0.20 (0.40)        0.84 (0.36)     <0.001 ***
hispanic      0.14 (0.35)        0.06 (0.24)      0.003 **
married       0.51 (0.50)        0.19 (0.39)     <0.001 ***
nodegree      0.60 (0.49)        0.71 (0.46)      0.009 **
re74       5619.24 (6788.75) 2095.57 (4886.62) <0.001 ***
re75       2466.48 (3292.00) 1532.06 (3219.25)  0.001 ***
re78       6984.17 (7294.16) 6349.14 (7867.40)  0.334
u74           0.26 (0.44)        0.71 (0.46)     <0.001 ***
u75           0.31 (0.46)        0.60 (0.49)     <0.001 ***
```

⇒ Covariates are NOT distributed almost equally across groups.

- What does this imbalance imply?

LaLonde (1986): (causal) effect of a job training program on earnings (VI)

Effect Evaluation 1: Difference of Means (Marginal)

```
> mean(lalonde$re78[lalonde$treatment==1]) - mean(lalonde$re78[lalonde$
  treatment==0])
[1] -635.0262
```

$\Rightarrow$ $\text{ATE} = -635.03$, with C.I. (95%): $[-1,967.24;\ 697.19]$

Effect Evaluation 2: Linear Regression ("Conditional")

```
> lalonde_fit <- lm(re78 ~ ., data=lalonde)
Estimate Std. Error    t value   Pr(>|t|)
 865.586    800.010      1.082      0.280
```

$\Rightarrow$ $\text{ATE} = 865.59$ (though also not significant)

- What if we use only the covariates `treat` and `age?` ("Partially Conditional")

```
> lalonde_fit2 <- lm(re78 ~ treatment + age, data=lalonde)
Estimate Std. Error    t value   Pr(>|t|)
-462.060    657.677     -0.703      0.483
```

- The estimated effects are NOT similar! Why? Should this always be the case?

Day 1

Part 2: Potential Outcomes (Rubin's Model)

What do We Mean by Causal Effect?

- Consider the statement:

  "My headache went away because I took an aspirin."

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- It means that

  "If I hadn't taken an aspirin, my headache wouldn't have gone away."

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  "If I hadn't taken an aspirin, my headache wouldn't have gone away."

- So we have the (necessary) implication

  "Taking aspiring" $\Rightarrow$ "Headache ending"

- Let $Y(a) \in \{0, 1\}$ be my "headache state" if we had been subject to intervention

  $$a \in \{\text{"Take Aspirin"}, \text{"Don't Take Aspirin"}\}$$

  and

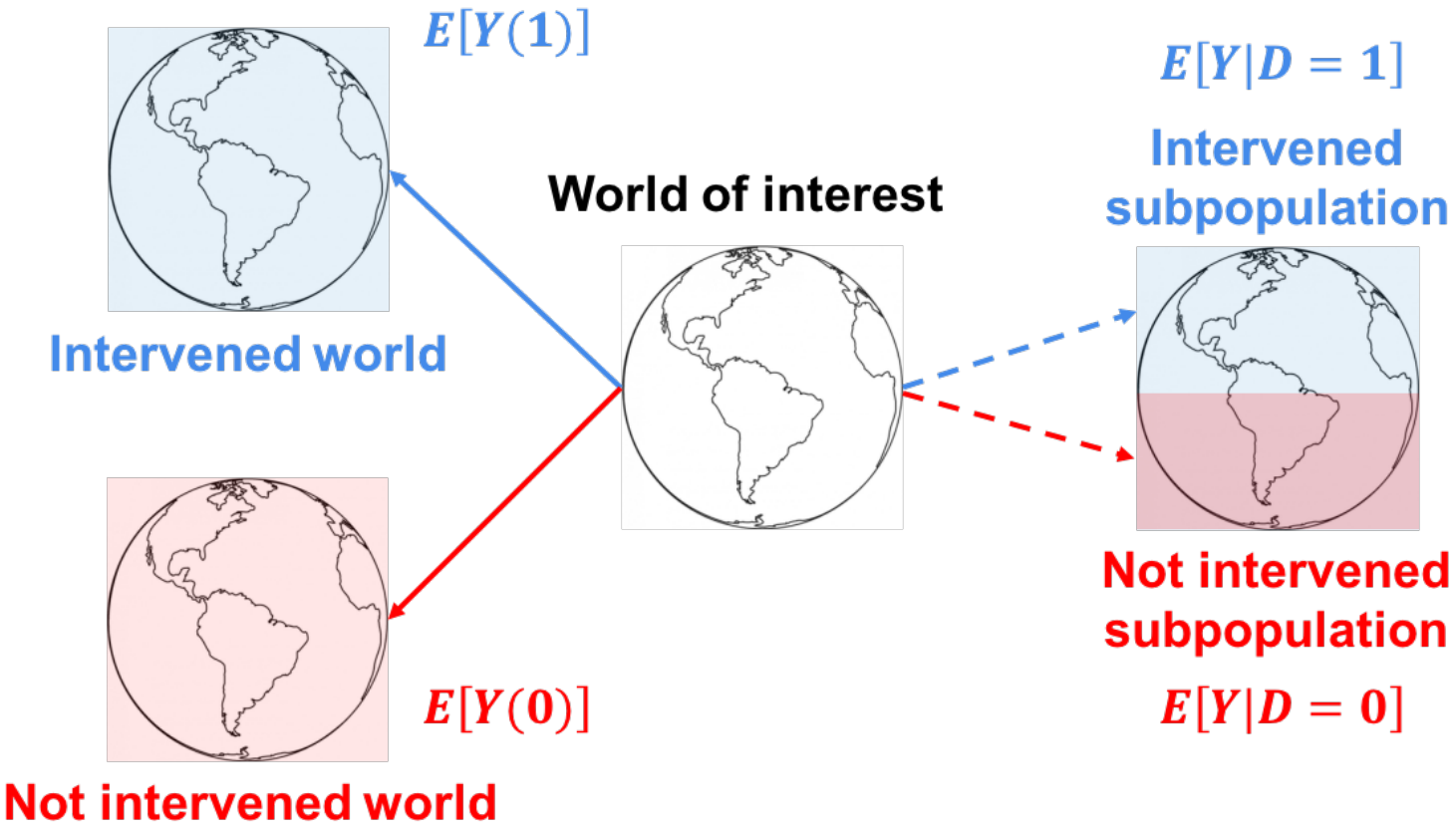
  $$Y(a) = 1 \Leftrightarrow \text{my headache goes away},$$

  then the INDIVIDUAL causal effect is

  $$Y(1) - Y(0).$$

Counterfactuals (1)

[A cause is] *"an object followed by another, and where all the objects, similar to the first, are followed by objects similar to the second. Or, in other words, where, if the first object had not been, the second never had existed." (Hume, 1748)*



Counterfactuals (2)

Basic idea: to compare potential outcomes under different states of treatment while keeping everything else invariant.

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$\Rightarrow$ at least one treatment assigment is counterfactual (and therefore unobservable)!

That is:

"If I took an aspirin, it is not possible that I didn't take an aspirin!"

But:

"I still want to know what would have happened if I had taken an aspirin!"

A Little More Formal...

Let:

A: treatment assignment (e.g., taking or not aspirin)

$$\begin{cases} A = 0 & \Rightarrow \quad \text{Control,} \\ A = 1 & \Rightarrow \quad \text{Treatment.} \end{cases}$$

Y: outcome (e.g., headache going away or not)

Remark: there is no need for A and/or Y to be binary...

Objective: **Evaluate the (causal) effect of A on Y.**

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Causal effect: any contrast between $Y(0)$ or $Y(1)$, e.g.,

$$Y(1) - Y(0) \quad \text{or} \quad \frac{Y(1)}{Y(0)}$$

Fundamental Problem of Causal Inference

THE POTENTIAL OUTCOMES $Y(0)$ AND $Y(1)$ CANNOT BOTH BE OBSERVED!

(Either you take aspirin or you don't.)

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"SOLUTION": causal inference at the "population level"? How?

- The population version of $Y(1) - Y(0)$ is the AVERAGE TREATMENT EFFECT (ATE):

$$\mathrm{ATE} = \mathrm{E}\left(Y(1) - Y(0)\right).$$

- How could we possibly estimate the ATE?

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- How could we possibly estimate the ATE?

NAIVE APPROACH: replace expectations by their empirical counterparts.

$$\widehat{\tau} = \frac{1}{n}\sum_{i=1}^{n} Y_i(1) - \frac{1}{n}\sum_{i=1}^{n} Y_i(0).$$

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$$\widehat{\tau} = \frac{1}{n}\sum_{i=1}^{n} Y_i(1) - \frac{1}{n}\sum_{i=1}^{n} Y_i(0).$$

- But that doesn't make any sense:

  Half of these variables are missing!!

i	A	$Y(0)$	$Y(1)$	Y
1	1	*NA*	y_1	y_1
2	0	y_2	*NA*	y_2
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
n	0	y_n	*NA*	y_n

Randomized Experiments

Def. Probabilistic (treatment/control) assignment mechanism whose distribution is known.

(Flip a coin. If 'Heads', then 'Treatment'. Otherwise, 'Control'.)

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2. The potential outcomes are independent of the treatment assignment:

$$Y(0), Y(1) \perp\!\!\!\perp A$$

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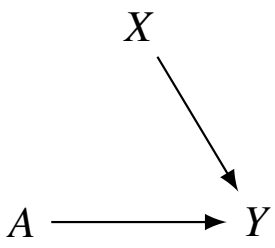
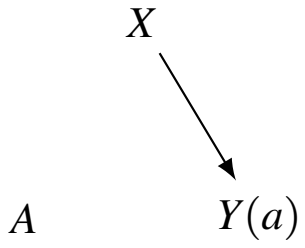


(a) Observed outcome



(b) Intervened outcome

Randomized Experiments

Def. Probabilistic (treatment/control) assignment mechanism whose distribution is known.

(Flip a coin. If 'Heads', then 'Treatment'. Otherwise, 'Control'.)

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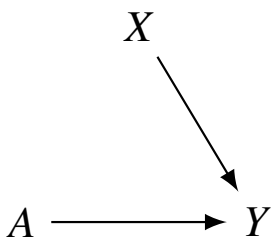

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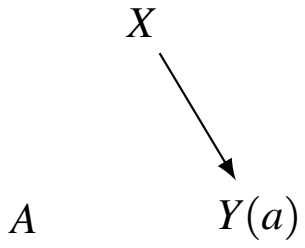


(b) Intervened outcome

$$\Rightarrow \mathrm{E}\, Y(a) = \mathrm{E}[Y|A = 1]$$

Day 1

Part 3A: Causal Effect (Basics)

Simulation: Randomized Experiment (I)

For $i = 1, \ldots, n$, let

$$A_i \sim \text{Bernoulli}(p) \quad \text{and} \quad Y_i = 0.25 + 1.5A_i + 0.3X_{i1} + 0.15X_{i2} + \varepsilon_i$$

with $A \perp\!\!\!\perp X_1, X_2$ and $\varepsilon_i \overset{\text{ind}}{\sim} \mathcal{N}(0, 0.04)$.

Simulation: Randomized Experiment (I)

For $i = 1, \ldots, n$, let

$$A_i \sim \text{Bernoulli}(p) \quad \text{and} \quad Y_i = 0.25 + 1.5A_i + 0.3X_{i1} + 0.15X_{i2} + \varepsilon_i$$

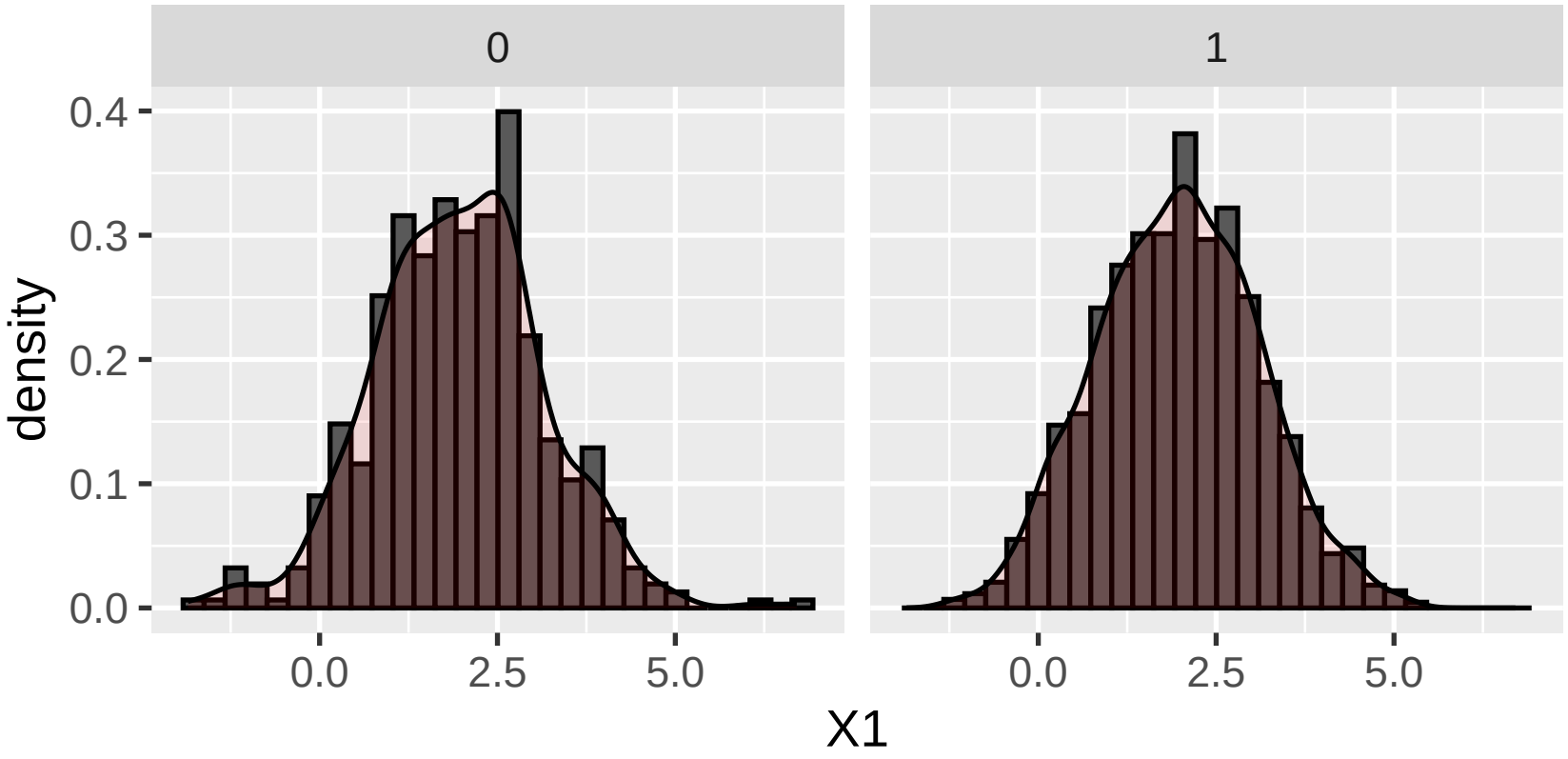
with $A \perp\!\!\!\perp X_1, X_2$ and $\varepsilon_i \overset{\text{ind}}{\sim} \mathcal{N}(0, 0.04)$.

```
> library(mvtnorm)
> library(ggplot2)
>
> # Sample size:
> n <- 2000
> # Treatment assignment:
> p <- .75
> A <- rbinom(n, 1, p)
> # Covariates:
> X <- rmvnorm(n, mean=c(2,-3), sigma=matrix(c(1.23,-.2,-.2,.93), nrow=2))
> # Outcome:
> tau <- 1.5 # Causal effect!
> Y <- 0.25 + tau*A + 0.3*X[,1] + 0.15*X[,2] + 0.2*rnorm(n)
> # Data frame:
> df <- data.frame(A,X,Y)
> head(df)
  A       X1        X2         Y
1 1 2.442715 -4.134884 1.6364982
2 1 2.233993 -2.301823 1.5449077
3 1 3.252955 -3.061028 2.1435520
4 1 3.330279 -3.186272 2.1622453
```

Simulation: Randomized Experiment (II)

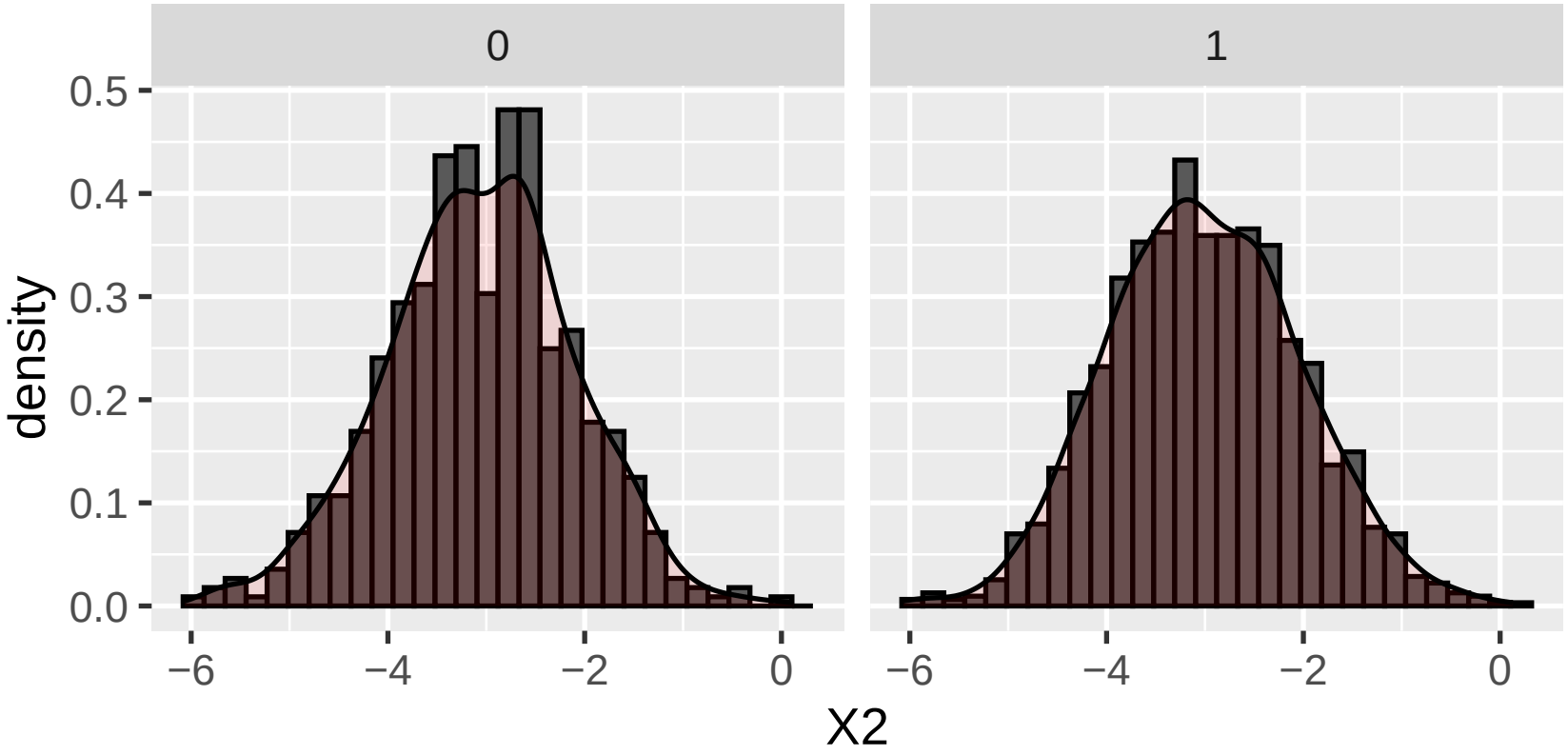
CHECKING COVARIATE BALANCE X_1:

```
> ggplot(df, aes(x=X1)) +
+   geom_histogram(aes(y = ..density..), color="black") +
+   geom_density(fill="red", alpha = 0.1) +
+   facet_wrap(~A)
```



CHECKING COVARIATE BALANCE X_2:

```
> ggplot(df, aes(x=X2)) +
+   geom_histogram(aes(y = ..density..), color="black") +
+   geom_density(fill="red", alpha = 0.1) +
+   facet_wrap(~A)
```



Simulation: Randomized Experiment (IV)

ESTIMATING THE CAUSAL EFFECT (DIFFERENCE OF MEANS):

```
> ## Estimating the causal effect
> # Difference of means
> mean(df$Y[df$A==1]) - mean(df$Y[df$A==0])
[1] 1.488934
> sqrt(var(df$Y[df$A==1]) + var(df$Y[df$A==0]))
[1] 0.5628033
```

$\Rightarrow$ ATE $= 1.49$ (0.56)

Simulation: Randomized Experiment (IV)

Estimating the causal effect (difference of means):

```
> ## Estimating the causal effect
> # Difference of means
> mean(df$Y[df$A==1]) - mean(df$Y[df$A==0])
[1] 1.488934
> sqrt(var(df$Y[df$A==1]) + var(df$Y[df$A==0]))
[1] 0.5628033
```

$\Rightarrow$ ATE $= 1.49$ (0.56)

Estimating the causal effect (regression):

```
> # Regression
> summary(lm(Y ~ ., data=df))

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.254552   0.017450   14.59   <2e-16 ***
A           1.482819   0.010233  144.91   <2e-16 ***
X1          0.299525   0.004029   74.35   <2e-16 ***
X2          0.145974   0.004805   30.38   <2e-16 ***
```

$\Rightarrow$ ATE $= 1.48$ (0.01)

Why Are These Estimates Correct? And SUTVA! (I)

QUESTION 1: Why $\mathrm{E}\, Y(a) = \mathrm{E}[Y|A = a]$ in randomized experiments?

ANSWER: remember first that (**in a randomized study**),

$$Y(0), Y(1) \perp\!\!\!\perp A \quad \Rightarrow \quad \mathrm{E}\, Y(a) = \mathrm{E}[Y(a)|A = a] \quad (\star)$$

QUESTION 2: How to transform $Y(a)$ into Y?

ANSWER: Stable Unit Treatment Value Assumption (SUTVA)

No unit is affected by assignments to other units, and there are no different versions of each treatment level leading to different outcomes.

Meaning (for binary interventions):

$$Y = AY(1) + (1 - A)Y(0)$$

Puting together the SUTVA and $(\star)$,

$$\Rightarrow \mathrm{E}\, Y(a) = \mathrm{E}[Y|A = a]$$

Why Are These Estimates Correct? And SUTVA! (II)

CONCLUSION: In randomized experiments with binary treatments, we always have

$$\mathrm{ATE} = \mathrm{E}[Y|A=1] - \mathrm{E}[Y|A=0]$$

REMARK: If the treatment has multiple levels, there will be multiple contrasts.

Example: Suppose the treatment is randomly assigned and that

$$Y = \alpha + \tau A + \beta^\top X + \varepsilon, \quad \mathrm{E}\,\varepsilon = 0.$$

Since this is a randomized experiment,

$$\begin{aligned}\mathrm{ATE} &= \mathrm{E}[Y|A=1] - \mathrm{E}[Y|A=0] \\ &= (\alpha + \tau \times 1 + \beta^\top \mathrm{E}\,X) - (\alpha + \tau \times 0 + \beta^\top \mathrm{E}\,X) = \tau.\end{aligned}$$

Caution: $\mathrm{E}[\varepsilon|A] = 0$ (or, more strongly, $\varepsilon \perp\!\!\!\perp A$)

$\Rightarrow$ ε does NOT confound the relation between A and Y!

Day 1

Part 3B: Causal Effect (Confounding)

Confounding

- $X_1, \ldots, X_p$ are CONFOUNDERS for the relationship between A and Y if

$$Y(a) \not\perp\!\!\!\perp A \quad \text{but} \quad Y(a) \perp\!\!\!\perp A|X$$

$$\begin{tikzcd} & X \arrow[dl] \arrow[dr] & \\ A \arrow[rr] & & Y \end{tikzcd}$$

(Confounded treatment and outcome)

$$\begin{tikzcd} & X \arrow[dl] \arrow[dr] & \\ A & & Y(a) \end{tikzcd}$$

(Dependent treatment and potential outcome)

$$\begin{tikzcd} & \boxed{X = x} \arrow[dl] \arrow[dr] & \\ A & & Y(a) \end{tikzcd}$$

(Conditionally independent treatment and potential outcome)

- In randomized experiments, there is no confounding.
- If the study is not randomized, it is said to be an OBSERVATIONAL STUDY.

Simpson's Paradox and Confounding (I): Observational Study on Kidney Stones

- A: **treatment** (open surgical procedure (1) vs. small puncture procedure (0));
- Y: **outcome** (success (1) vs. failure (0)).

Simpson's Paradox and Confounding (I): Observational Study on Kidney Stones

- A: **treatment** (open surgical procedure (1) vs. small puncture procedure (0));
- Y: **outcome** (success (1) vs. failure (0)).

DATA:

	$Y = 0$	$Y = 1$
$A = 0$	61	289
$A = 1$	77	273

Simpson's Paradox and Confounding (I): Observational Study on Kidney Stones

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RISK DIFFERENCE:

	$Y = 0$	$Y = 1$
$A = 0$	Y_{00}	Y_{01}
$A = 1$	Y_{10}	Y_{11}

$$\Rightarrow \quad RD = \frac{Y_{11}}{Y_{10} + Y_{11}} - \frac{Y_{01}}{Y_{00} + Y_{01}}.$$

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$A=1$	Y_{10}	Y_{11}

$$\Rightarrow \quad RD = \frac{Y_{11}}{Y_{10}+Y_{11}} - \frac{Y_{01}}{Y_{00}+Y_{01}}.$$

ESTIMATED RISK DIFFERENCE:

$$\widehat{RD} = \underbrace{\frac{273}{77+273}}_{\text{Surgery}} - \underbrace{\frac{289}{61+289}}_{\text{Puncture}} = -5\%$$

Simpson's Paradox and Confounding (II): Observational Study on Kidney Stones

SUBGROUPS: stratification by severity

- S: **subgroup** (larger stones (1) vs. smaller stones (0)).

Simpson's Paradox and Confounding (II): Observational Study on Kidney Stones

SUBGROUPS: stratification by severity

- S: **subgroup** (larger stones (1) vs. smaller stones (0)).

Smaller Stones

	$Y=0$	$Y=1$	Total
$A=0$	36	234	270
$A=1$	6	81	87
Total	42	315	**357**

Larger Stones

	$Y=0$	$Y=1$	Total
$A=0$	25	55	80
$A=1$	71	192	263
Total	96	247	**343**

Simpson's Paradox and Confounding (II): Observational Study on Kidney Stones

SUBGROUPS: stratification by severity

- S: **subgroup** (larger stones (1) vs. smaller stones (0)).

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	$Y=0$	$Y=1$	Total
$A=0$	36	234	270
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Total	42	315	**357**

Larger Stones

	$Y=0$	$Y=1$	Total
$A=0$	25	55	80
$A=1$	71	192	263
Total	96	247	**343**

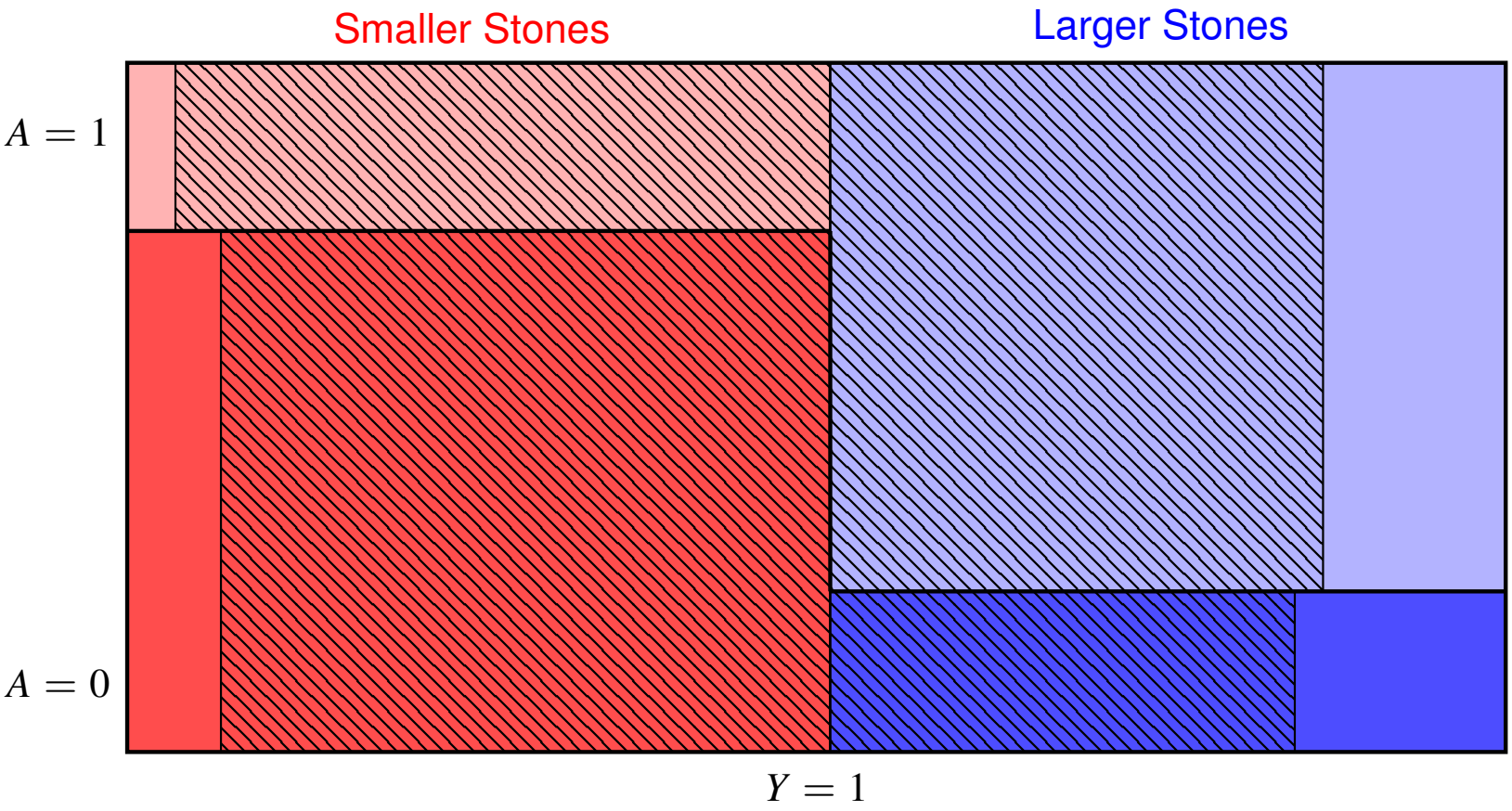
ESTIMATED RISK DIFFERENCES:

$$\widehat{RD}_{\text{smaller}} = \underbrace{\frac{81}{6+81}}_{\text{Surgery}} - \underbrace{\frac{234}{36+234}}_{\text{Puncture}} = 6\%$$

$$\widehat{RD}_{\text{larger}} = \underbrace{\frac{192}{71+192}}_{\text{Surgery}} - \underbrace{\frac{55}{25+55}}_{\text{Puncture}} = 4\%$$

Simpson's Paradox and Confounding (III): Observational Study on Kidney Stones

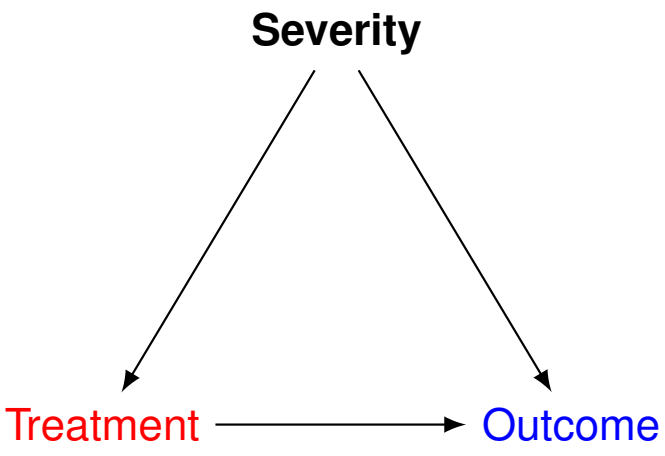
EXPLANATION (I):



Simpson's Paradox and Confounding (IV): Observational Study on Kidney Stones

EXPLANATION (II):

- **Severity** affects both treatment assignment and outcome



- Larger stones $\Rightarrow$ riskier (more $Y = 0$).
- Larger stones $\Rightarrow$ surgical procedure (more $A = 1$).
- Disregarding severity $\Rightarrow$ unbalanced sample!

g-Formula

QUESTION: how to deal with unbalanced samples in observational studies?

1st STRATEGY: g-FORMULA

$$\begin{aligned}
\mathrm{E}\, Y(a) &= \mathrm{E}^X\, \mathrm{E}\left(Y(a)|X\right) \quad \text{(Law of Iterated Expectations)} \\
&= \mathrm{E}^X\, \mathrm{E}\left(Y(a)|A=a, X\right) \quad \text{(Ignorability, or } Y(a) \perp\!\!\!\perp A|X\text{)} \\
&= \mathrm{E}^X\, \mathrm{E}\left(Y|A=a, X\right) \quad \text{(Consistency, or } Y(a)=Y\text{, if } A=a\text{)}
\end{aligned}$$

Hence, the g-formula:

$$\mathrm{E}\left(Y(1)-Y(0)\right) = \mathrm{E}^X\left[\mathrm{E}(Y|A=1,X) - \mathrm{E}(Y|A=0,X)\right]$$

Conditionally on $X=x$, we have

$$\mathrm{E}\left(Y(1)-Y(0)|X=x\right) = \mathrm{E}(Y|A=1,X=x) - \mathrm{E}(Y|A=0,X=x)$$

Simpson's Paradox and Confounding (IV): Observational Study on Kidney Stones

Adjusting for the Confounding:

Pooled Sample

	$Y=0$	$Y=1$
$A=0$	61	289
$A=1$	77	273

Smaller Stones $(S=0)$

	$Y=0$	$Y=1$
$A=0$	36	234
$A=1$	6	81

Larger Stones $(S=1)$

	$Y=0$	$Y=1$
$A=0$	25	55
$A=1$	71	192

From g-formula:

$$
\begin{aligned}
\mathrm{E}\left(Y(1)-Y(0)\right) &= \mathrm{E}^{X}\left[\mathrm{E}(Y|A=1,S)-\mathrm{E}(Y|A=0,S)\right] \\
&= \left(\mathrm{E}(Y|A=1,S=1)-\mathrm{E}(Y|A=0,S=1)\right)\mathrm{P}(S=1) \\
&\quad + \left(\mathrm{E}(Y|A=1,X=0)-\mathrm{E}(Y|A=0,S=0)\right)\mathrm{P}(S=0) \\
&= \left(\frac{192}{71+192}-\frac{55}{25+55}\right)\times\frac{343}{700}+\left(\frac{81}{6+81}-\frac{234}{36+234}\right)\times\frac{357}{700} \\
&= (0.7300-0.6875)\times 0.49+(0.9310-0.8667)\times 0.51
\end{aligned}
$$

$$\Rightarrow \mathrm{E}\left(Y(1)-Y(0)\right) \approx 5.36\%$$

Would the Simpson's Paradox Happen in a Randomized Study?

In a randomized study,

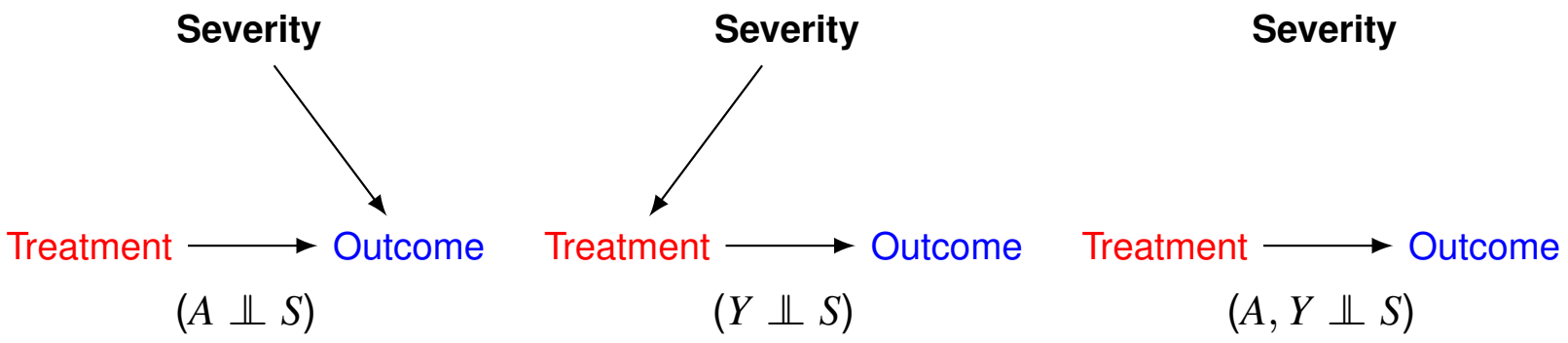
$$Y(a) \perp\!\!\!\perp A,$$

so that

$$\begin{aligned} \mathrm{E}\left(Y(1) - Y(0)\right) &= \mathrm{E}\left(Y(1)|A = 1\right) - \mathrm{E}\left(Y(0)|A = 0\right) \\ &= \mathrm{E}\left(Y|A = 1\right) - \mathrm{E}\left(Y|A = 0\right) \end{aligned}$$

$\Rightarrow$ the ATE is just the difference of the expectations conditioned only on A.

- In a randomized study, one of the following must hold:



Day 1

Part 3C: Causal Effect (Regression)

Regression Models in Causal Inference

- Consider again the (conditional) ATE:

$$\mathrm{ATE}(x) = \mathrm{E}\left(Y(1) - Y(0) | X = x\right).$$

- g-formula says that:

$$\mathrm{ATE}(x) = \mathrm{E}(Y|A = 1, X = x) - \mathrm{E}(Y|A = 0, X = x)$$

- This suggests estimating the ATE using

$$\widehat{\mathrm{ATE}}(x) = \widehat{\mathrm{E}}(Y|A = 1, X = x) - \widehat{\mathrm{E}}(Y|A = 0, X = x)$$

where $\widehat{\mathrm{E}}(Y|A = a, X = x)$ is a regression model for $\mathrm{E}(Y|A = a, X = x)$.

Linear Models in Causal Inference

ASSUMPTION 1:

$$\underbrace{Y = \tau A + \alpha^\top X + \varepsilon}_{\text{structural equation}},$$

where $\mathrm{E}(\varepsilon|A, X) = 0$.

ASSUMPTION 2: no unobserved confounding!

- Using the g-formula:

$$\mathrm{ATE}(x) = \left(\tau + \alpha^\top x\right) - \left(\alpha^\top x\right) = \tau.$$

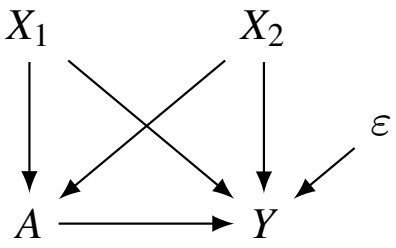
REMARK: $\mathrm{ATE}(x) \equiv \mathrm{ATE}$, for all x

$\Rightarrow$ Assumption 1 makes sense only in the absence of heterogeneity!

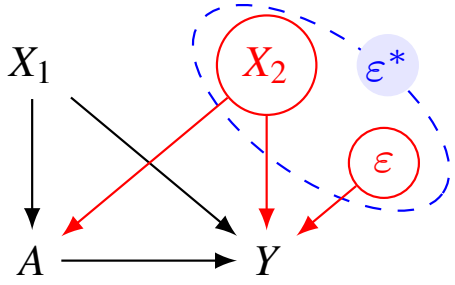
QUESTION: what happens if Assumption 2 fails?

Linear Models in Causal Inference: Endogeneity (I)

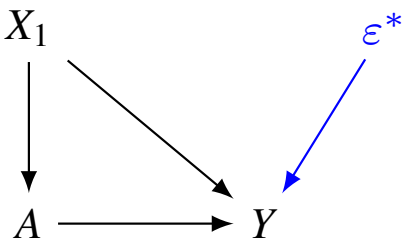
OMMITED VARIABLES: $X^\top = (X_1^\top, X_2^\top)$, but X_2 is unobserved.



(Exogenous error)



(Unobserved confounding)



(Endogenous "error")

PROBLEM (ENDOGENEITY): $\mathrm{Cov}(A, \varepsilon) \neq 0 \quad \Rightarrow \quad Y(a) \not\perp\!\!\!\perp A | X_1$

CONSEQUENCE: given the model $Y = \tau A + \alpha_1^\top X_1 + \varepsilon^*$,

$$\Rightarrow \mathrm{E}(Y|A = a, X_1) = \tau a + \alpha_1^\top X_1 + \underbrace{\mathrm{E}(\varepsilon^*|A = a, X_1)}_{=\alpha_2^\top \mathrm{E}(X_2|A=a,X_1)},$$

so that

$$\mathrm{E}(Y|A = 1, X_1) - \mathrm{E}(Y|A = 0, X_1) = \tau + \underbrace{\alpha_2^\top \left(\mathrm{E}(X_2|A = 1, X_1) - \mathrm{E}(X_2|A = 0, X_1)\right)}_{\text{Omitted Variables Bias}}$$

EXAMPLE:

$$Y = 0.75 + A + 1.4X_1 + 5X_2 + \varepsilon$$

$$\mathrm{P}(A = 1|X = x) = \frac{\exp(\eta(x))}{1 + \eta(x)}, \quad \eta(x) = 0.8X_1 - 0.5X_2$$

```
> library(mvtnorm)
> set.seed(1234)
> # Sample size
> n <- 40
> # Effect
> tau <- 1
> # Covariates simulation
> X <- rmvnorm(n, mean=c(1,2),
+              sigma=matrix(c( .750, -.375,
+                              -.375,   1),
+                           nrow=2))
> # Treatment assignment simulation
> coef_A <- c(.3,-.5)
> A <- X %*% coef_A + rlogis(n)
> A <- ifelse(A<0, 0, 1)
> # Outcome simulation
> coef_Y <- c(1.4, 5)
> Y <- .75 + tau*A + X%*%coef_Y + .75*rnorm(n)
> # Data
> dataset <- data.frame(A, X, Y)
```

$\Rightarrow$

```
> dataset
   A     X1    X2    Y
1  0 -0.072  2.52 13.6
2  1  2.396 -0.52  2.2
3  0  1.257  2.41 14.6
4  0  0.629  1.58  9.2
5  0  0.709  1.25  7.3
6  1  0.804  1.12  8.6
7  0  0.334  2.22 11.7
8  0  1.830  1.69 11.9
9  1  0.758  1.21  9.1
10 0 -0.202  4.54 23.1
11 1  1.214  1.49 10.8
12 0  0.535  2.54 13.7
13 1  0.715  0.73  5.5
14 0  1.694  0.88  8.2
15 0  1.180  1.09  7.9
16 1  2.025  1.31 11.7
17 0  0.507  1.66  8.8
18 0 -0.130  1.19  6.7
19 0 -0.557  1.14  6.2
```

Linear Models in Causal Inference: Endogeneity (III)

Fit 1 (complete data):

```
> # Complete data fit:
> fit_comp <- lm(Y ~ ., data=dataset)
> summary(fit_comp)

            Estimate Std. Error t value Pr(>|t|)
(Intercept)    1.041      0.307    3.39   0.0017 **
A              1.069      0.225    4.76  3.1e-05 ***
X1             1.148      0.136    8.45  4.5e-10 ***
X2             4.900      0.116   42.37  < 2e-16 ***
```

$$\widehat{\tau} = 1.069$$

Fit 2 (partial data): X_2 removed

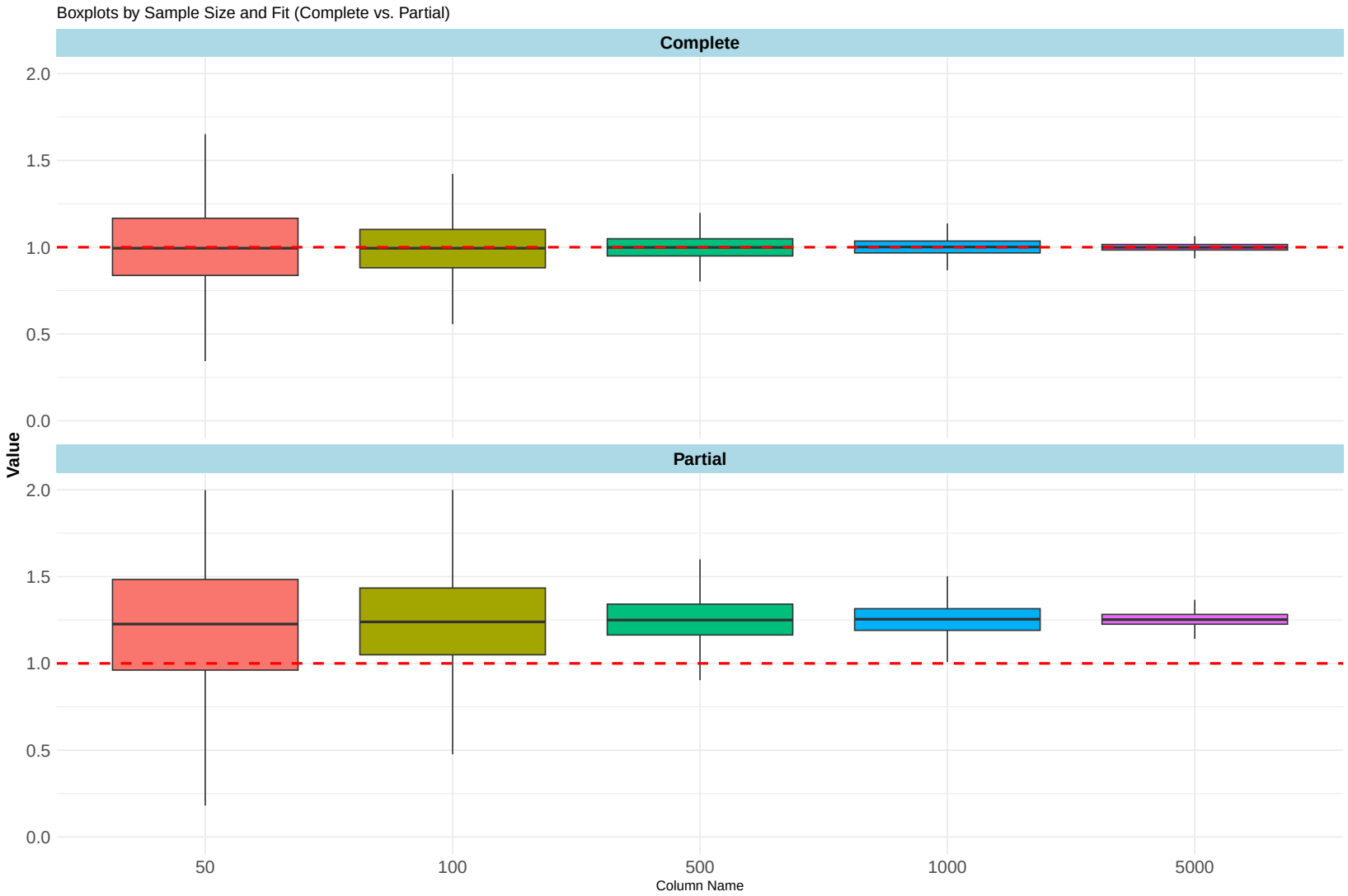
```
> # Partial data fit (X2 removed)
> fit_part <- lm(Y ~ A + X2, data=dataset)
> summary(fit_part)

            Estimate Std. Error t value Pr(>|t|)
(Intercept)    2.460      0.438    5.62    2e-06 ***
A              1.564      0.369    4.24  0.00014 ***
X2             4.559      0.185   24.69  < 2e-16 ***
```

$$\widehat{\tau} = 1.564$$

(50% overestimated!)

Does sample size affect the result? ($G = 3.000$ simulations)



Linear Models under Heterogeneity (I)

HETEROGENEITY: treatment effect varies across individuals.

INTERACTION (OR TREATMENT MODIFICATION): if $X = (S, W)$, then

$$Y = \tau A + \delta^\top (AS) + \alpha^\top X + \varepsilon$$

CONDITIONAL ATE:

$$\begin{aligned}
\mathrm{ATE}(s, w) &= \mathrm{E}\left(Y(1) - Y(0) | S = s, W = w\right) \\
&= \mathrm{E}\left(Y(1) | S = s, W = w\right) - \mathrm{E}\left(Y(0) | S = s, W = w\right) \\
&= \mathrm{E}\left(Y(1) | A = 1, S = s, W = w\right) - \mathrm{E}\left(Y(0) | A = 0, S = s, W = w\right) \\
&= \mathrm{E}\left(Y | A = 1, S = s, W = w\right) - \mathrm{E}\left(Y | A = 0, S = s, W = w\right)
\end{aligned}$$

$$\Rightarrow \quad \mathrm{ATE}(s) = (\tau + \delta^\top s + \alpha^\top X) - (\alpha^\top X) = \tau + \delta^\top s.$$

EXAMPLE: Let S be binary (e.g., gender or college degree)

$$\mathrm{ATE}(x_1) = \begin{cases} \tau, & \text{se } s = 0, \\ \tau + \delta, & \text{se } s = 1. \end{cases}$$

Linear Models under Heterogeneity (II)

"MARGINAL" EFFECT: $\mathrm{E}(Y(1) - Y(0)|W) = ?$

- Assuming S binary (for simplicity),

$$\mathrm{ATE} = \mathrm{E}(Y(1) - Y(0)) = \mathrm{E}^X \, \mathrm{E}\left(Y(1) - Y(0)|S, W\right) = \mathrm{E}^X \, \mathrm{ATE}(S, W)$$

$$\Rightarrow \quad \mathrm{ATE} = \mathrm{E}^W \left[\mathrm{ATE}(0, W) \, \mathrm{P}(S = 0|W) + \mathrm{ATE}(1, W) \, \mathrm{P}(S = 1|W)\right].$$

- If $Y = \tau A + \delta AS + \alpha^\top X + \varepsilon$, then

$$\begin{aligned}\mathrm{ATE} &= \mathrm{E}^W \left[\tau \, \mathrm{P}(S = 0|W) + (\tau + \delta) \, \mathrm{P}(S = 1|W)\right] \\ &= \tau + \delta \, \mathrm{E}^W \, \mathrm{P}(S = 1|W) \\ &= \tau + \delta \, \mathrm{P}(S = 1).\end{aligned}$$

Model Misspecification (I)

FACT: model misspecification can bias the estimate of the causal effect, EVEN IF all confounding factors are observed...

EXAMPLE:

- Structural equations:

$$Y = 0.75 + A + 4|X| + 0.5\varepsilon$$

$$\mathrm{P}(A = 1|X = x) = \frac{\exp(\eta(x))}{1 + \eta(x)}, \quad \eta(x) = -0.5 + 2X$$

- Misspecified model:

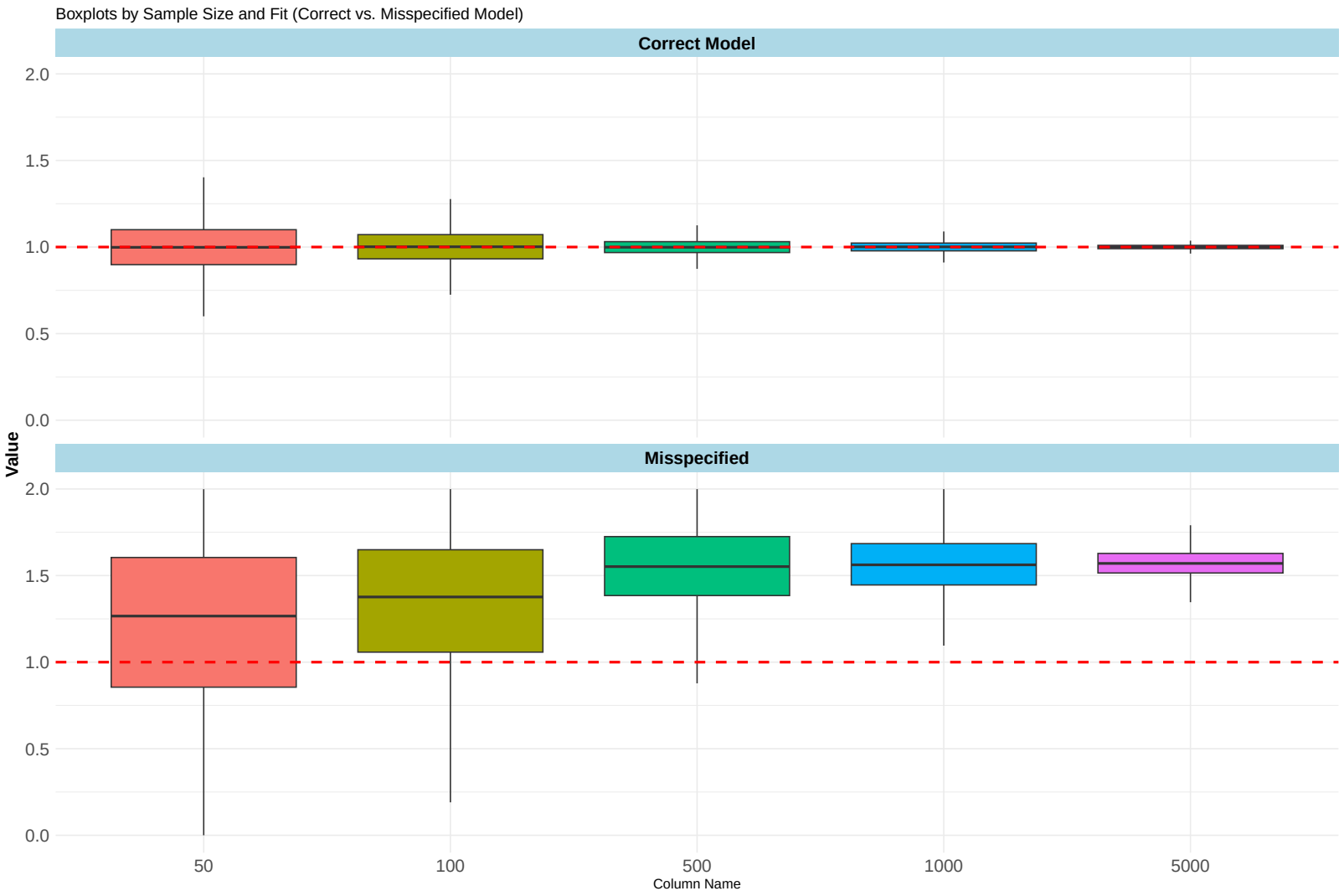
$$Y = \alpha_0 + \tau A + \alpha X + \varepsilon. \quad (\star)$$

- Biased causal effect:

$$\tau = \mathrm{E}\left(Y|A = 1, X\right) - \mathrm{E}\left(Y|A = 0, X\right).$$

- Biased effect estimate: $\widehat{\tau}$ (from model $(\star)$).

Is model misspecification a source of bias for the causal effect estimate? ($G = 3.000$ simulations)



Day 2

Part 4A: Propensity Scores

Another Strategy: Propensity Scores

UNDERLYING IDEA: causal inference is about balancing.

$\Rightarrow$ If we can balance the sample, we can infer the causal effect...

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PROPENSITY SCORE (PS):

$$\pi(X) = \mathrm{P}(A = 1|X).$$

Another Strategy: Propensity Scores

UNDERLYING IDEA: causal inference is about balancing.

$\Rightarrow$ If we can balance the sample, we can infer the causal effect...

PROPENSITY SCORE (PS):

$$\pi(X) = \mathrm{P}(A = 1|X).$$

	$A = 0$	$A = 1$
$X = 1$:	$1 - \pi(1)$	$\pi(1)$
$X = 2$:	$1 - \pi(2)$	$\pi(2)$
$X = 3$:	$1 - \pi(3)$	$\pi(3)$
$X = 4$:	$1 - \pi(4)$	$\pi(4)$

(Randomized Study: $A \perp\!\!\!\perp X$)

Another Strategy: Propensity Scores

UNDERLYING IDEA: causal inference is about balancing.

$\Rightarrow$ If we can balance the sample, we can infer the causal effect...

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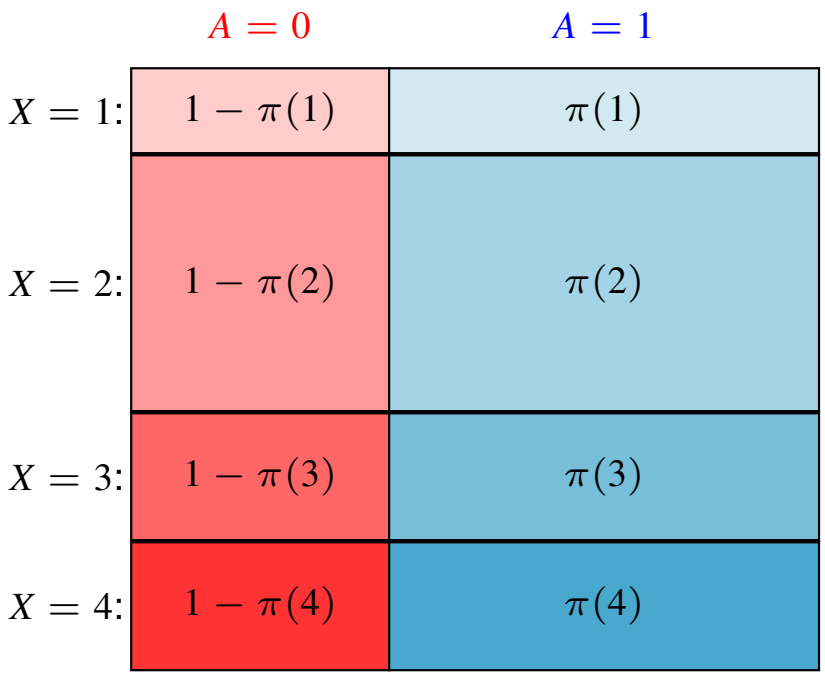
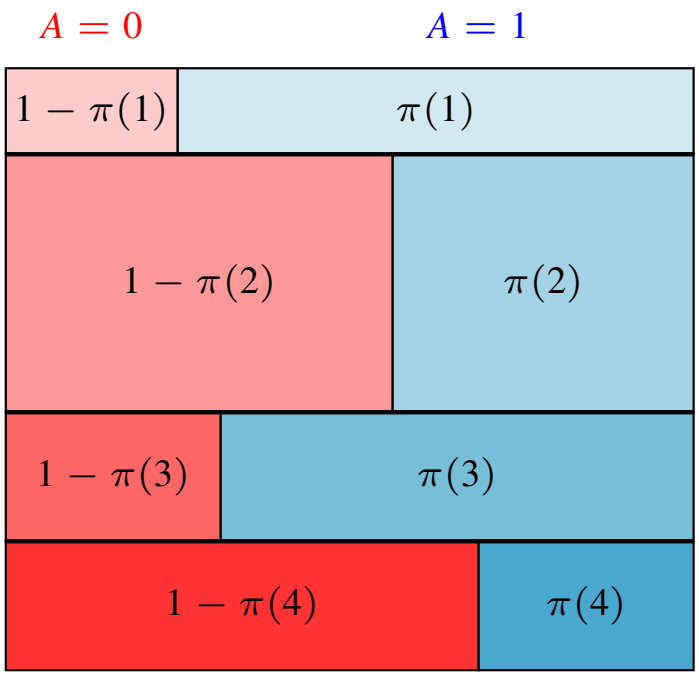


(Randomized Study: $A \perp\!\!\!\perp X$)



(Observational Study: $A \not\perp\!\!\!\perp X$)

Balancing Property

THEOREM. The PS is a balancing score, i.e.,

$$A \perp\!\!\!\perp X | \pi(X).$$

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PROOF.

- $\mathrm{P}(A = 1|X, \pi(X)) = \mathrm{P}(A = 1|X) = \pi(X)$

Balancing Property

THEOREM. The PS is a balancing score, i.e.,

$$A \perp\!\!\!\perp X | \pi(X).$$

PROOF.

- $\mathrm{P}(A = 1|X, \pi(X)) = \mathrm{P}(A = 1|X) = \pi(X)$
- On the other hand,

$$\begin{aligned}
\mathrm{P}(A = 1|\pi(X)) &= \mathrm{E}\left(A|\pi(X)\right) \\
&= \mathrm{E}\left[\mathrm{E}\left(A|X, \pi(X)\right)|\pi(X)\right] \\
&= \mathrm{E}\left[\pi(X)|\pi(X)\right] \\
&= \pi(X)
\end{aligned}$$

$\therefore$ $\mathrm{P}(A = 1|X, \pi(X)) = \mathrm{P}(A = 1|\pi(X))$

Balancing Property

THEOREM. The PS is a balancing score, i.e.,

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PROOF.

- $\mathrm{P}(A = 1|X, \pi(X)) = \mathrm{P}(A = 1|X) = \pi(X)$
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\mathrm{P}(A = 1|\pi(X)) &= \mathrm{E}\left(A|\pi(X)\right) \\
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&= \mathrm{E}\left[\pi(X)|\pi(X)\right] \\
&= \pi(X)
\end{aligned}$$

$\therefore$ $\mathrm{P}(A = 1|X, \pi(X)) = \mathrm{P}(A = 1|\pi(X))$

$\Rightarrow$ $A \perp\!\!\!\perp X|\pi(X)$

Balancing Scores and Unconfoundedness

THEOREM. If $b(X)$ is balancing score and $A \perp\!\!\!\perp Y(0), Y(1)|X$, then

$$A \perp\!\!\!\perp Y(0), Y(1)|b(X).$$

Balancing Scores and Unconfoundedness

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PROOF.

- Note that:

$$\begin{aligned} \mathrm{P}(A = 1|Y(0), Y(1), b(X)) &= \mathrm{E}\left(A|Y(0), Y(1), b(X)\right) \\ &= \mathrm{E}\left[\mathrm{E}\left(A|Y(0), Y(1), X, b(X)\right) \middle| Y(0), Y(1), b(X)\right] \end{aligned}$$

Balancing Scores and Unconfoundedness

THEOREM. If $b(X)$ is balancing score and $A \perp\!\!\!\perp Y(0), Y(1)|X$, then

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- $A \perp\!\!\!\perp Y(0), Y(1)|X$ implies

$$\mathrm{P}(A = 1|Y(0), Y(1), b(X)) = \mathrm{E}\left[\mathrm{E}\left(A|X, b(X)\right) \middle| Y(0), Y(1), b(X)\right]$$

Balancing Scores and Unconfoundedness

THEOREM. If $b(X)$ is balancing score and $A \perp\!\!\!\perp Y(0), Y(1)|X$, then

$$A \perp\!\!\!\perp Y(0), Y(1)|b(X).$$

PROOF.

- Note that:

$$\begin{aligned} \mathrm{P}(A = 1|Y(0), Y(1), b(X)) &= \mathrm{E}\left(A|Y(0), Y(1), b(X)\right) \\ &= \mathrm{E}\left[\mathrm{E}\left(A|Y(0), Y(1), X, b(X)\right) \middle| Y(0), Y(1), b(X)\right] \end{aligned}$$

- $A \perp\!\!\!\perp Y(0), Y(1)|X$ implies

$$\mathrm{P}(A = 1|Y(0), Y(1), b(X)) = \mathrm{E}\left[\mathrm{E}\left(A|X, b(X)\right) \middle| Y(0), Y(1), b(X)\right]$$

- $b(X)$ balancing score implies

$$\mathrm{P}(A = 1|Y(0), Y(1), b(X)) = \mathrm{E}\left[\mathrm{E}\left(A|b(X)\right) \middle| Y(0), Y(1), b(X)\right] = \mathrm{E}\left(A|b(X)\right)$$

Balancing Scores and Unconfoundedness

THEOREM. If $b(X)$ is balancing score and $A \perp\!\!\!\perp Y(0), Y(1)|X$, then

$$A \perp\!\!\!\perp Y(0), Y(1)|b(X).$$

PROOF.

- Note that:

$$\begin{aligned} \mathrm{P}(A = 1|Y(0), Y(1), b(X)) &= \mathrm{E}\left(A|Y(0), Y(1), b(X)\right) \\ &= \mathrm{E}\left[\mathrm{E}\left(A|Y(0), Y(1), X, b(X)\right) \middle| Y(0), Y(1), b(X)\right] \end{aligned}$$

- $A \perp\!\!\!\perp Y(0), Y(1)|X$ implies

$$\mathrm{P}(A = 1|Y(0), Y(1), b(X)) = \mathrm{E}\left[\mathrm{E}\left(A|X, b(X)\right) \middle| Y(0), Y(1), b(X)\right]$$

- $b(X)$ balancing score implies

$$\mathrm{P}(A = 1|Y(0), Y(1), b(X)) = \mathrm{E}\left[\mathrm{E}\left(A|b(X)\right) \middle| Y(0), Y(1), b(X)\right] = \mathrm{E}\left(A|b(X)\right)$$

$\Rightarrow$ $\mathrm{P}(A = 1|Y(0), Y(1), b(X)) = \mathrm{P}(A = 1|b(X)) \qquad \therefore \qquad A \perp\!\!\!\perp Y(0), Y(1)|b(X)$

All Covariates vs. Propensity Scores (I)

FIT 1 (USING $X = (X_1, X_2)$):

```
> # Inference using X:
> fit_X <- lm(Y ~ ., data=dataset)
> summary(fit_X)

            Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.8078     0.0356    22.7   <2e-16 ***
A             0.9832     0.0235    41.8   <2e-16 ***
X1            1.3694     0.0138    99.4   <2e-16 ***
X2            4.9845     0.0121   413.1   <2e-16 ***
```

$$\widehat{\tau} = 0.9832$$

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(Intercept)   0.8078     0.0356    22.7   <2e-16 ***
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X1            1.3694     0.0138    99.4   <2e-16 ***
X2            4.9845     0.0121   413.1   <2e-16 ***
```

$\widehat{\tau} = 0.9832$

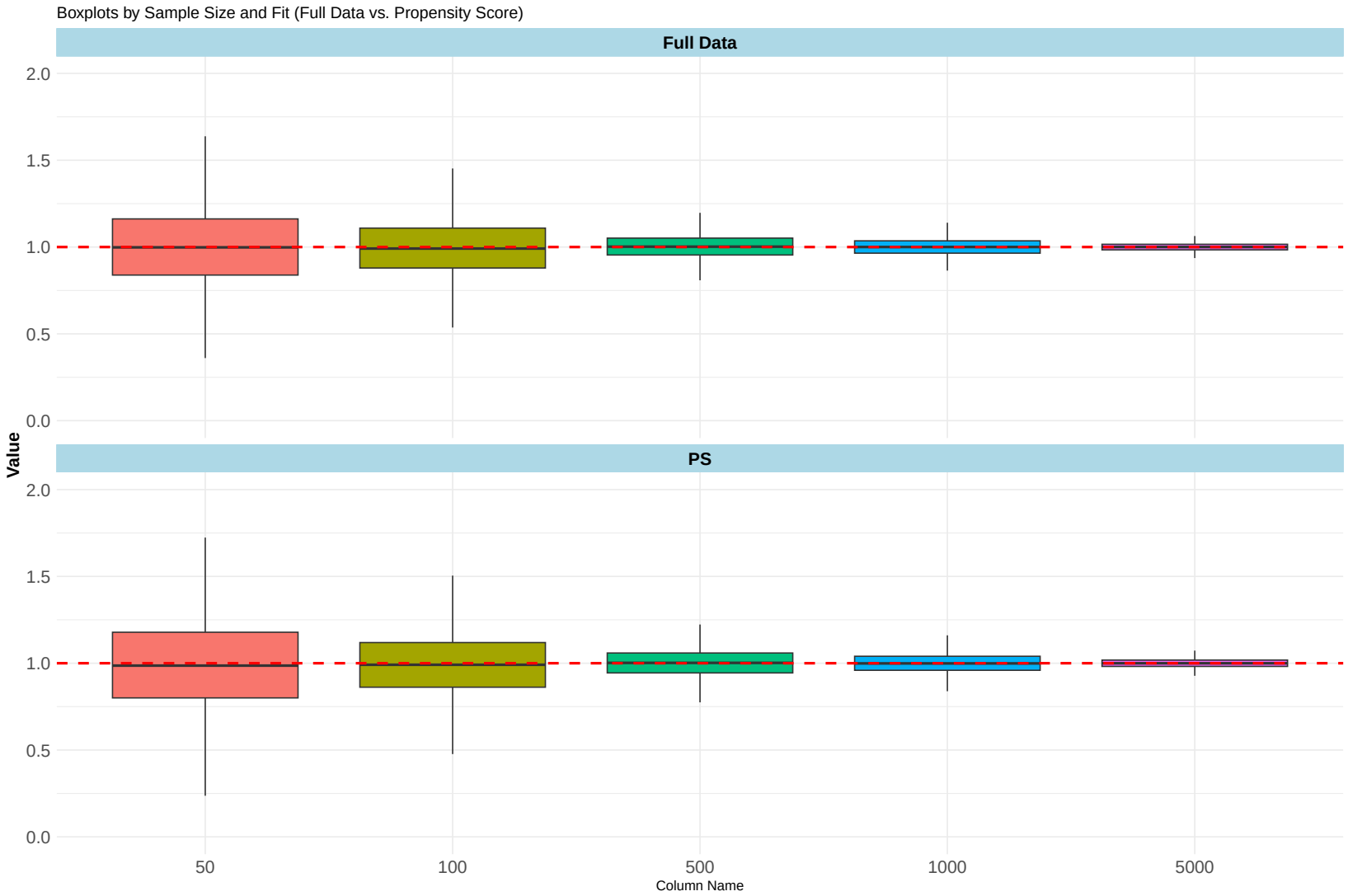
FIT 2 (USING $\widehat{\pi}(X)$)

```
> # Propensity score estimate:
> PS_fit <- glm(A ~ X1 + X2, data=dataset, family="binomial
     ")
> PS <- as.numeric(predict(PS_fit, type="response"))
> # Inference using the PS:
> fit_PS <- lm(Y ~ A + PS, data=dataset)
> summary(fit_PS)
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  21.4408     0.1155   185.6   <2e-16 ***
A             0.9838     0.0922    10.7   <2e-16 ***
PS          -26.9823     0.3277   -82.3   <2e-16 ***
```

$\widehat{\tau} = 0.9838$

All Covariates vs. Propensity Scores (II)

Does the use of (only) the PS's affect the result? ($G = 3.000$ simulations)

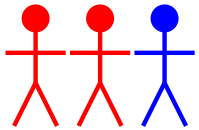


Day 2

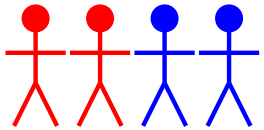
Part 4B: Propensity Scores (Inverse Probability Weighting)

Inverse Probability Weights (I)

ORIGINAL (UNBALANCED) SAMPLE:

$A = 1$: 

$$\begin{cases} \pi(\text{red}) = 1/2 \\ \pi(\text{blue}) = 1/3 \end{cases}$$

$A = 0$: 

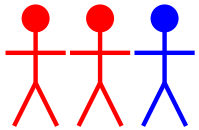
$$\begin{cases} 1 - \pi(\text{red}) = 1/2 \\ 1 - \pi(\text{blue}) = 2/3 \end{cases}$$

$$\mathrm{P}(\text{red}) = 4/7$$

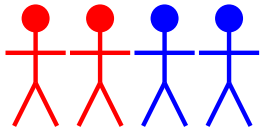
$$\mathrm{P}(\text{red}|A = 1) = 2/3$$
$$\mathrm{P}(\text{red}|A = 0) = 1/2$$

Inverse Probability Weights (I)

ORIGINAL (UNBALANCED) SAMPLE:

$A = 1$: 

$$\begin{cases} \pi(\text{red}) = 1/2 \\ \pi(\text{blue}) = 1/3 \end{cases}$$

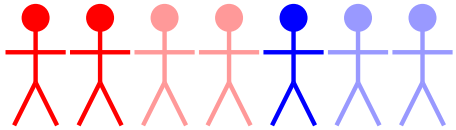
$A = 0$: 

$$\begin{cases} 1 - \pi(\text{red}) = 1/2 \\ 1 - \pi(\text{blue}) = 2/3 \end{cases}$$

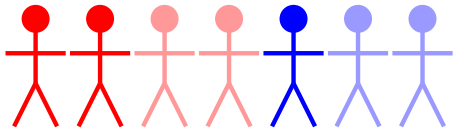
$$\mathrm{P}(\text{red}) = 4/7$$

$$\mathrm{P}(\text{red}|A = 1) = 2/3$$
$$\mathrm{P}(\text{red}|A = 0) = 1/2$$

AUGMENTED (BALANCED) SAMPLE:

$A = 1$: 

$$\begin{cases} \frac{1}{\pi(\text{red})} = 2 \\ \frac{1}{\pi(\text{blue})} = 3 \end{cases}$$

$A = 0$: 

$$\begin{cases} \frac{1}{1-\pi(\text{red})} = 2 \\ \frac{1}{1-\pi(\text{blue})} = 3/2 \end{cases}$$

$$\mathrm{P}(\text{red}) = 4/7$$

$$\mathrm{P}(\text{red}|A = 1) = 4/7$$
$$\mathrm{P}(\text{red}|A = 0) = 4/7$$

Inverse Probability Weights (II)

DEF.: For $A \in \{0, 1\}$, the IPW is given by

$$W \equiv W(X) = \frac{A}{\pi(X)} + \frac{1 - A}{1 - \pi(X)}$$

Inverse Probability Weights (II)

DEF.: For $A \in \{0, 1\}$, the IPW is given by

$$W \equiv W(X) = \frac{A}{\pi(X)} + \frac{1 - A}{1 - \pi(X)}$$

THEOREM. If $A \in \{0, 1\}$ and

(A1): (CONSISTENCY) $Y = AY(1) + (1 - A)Y(0)$;

(A2): (POSITIVITY) $0 < \pi(X) < 1$;

(A3): (IGNORABILITY) $Y(0), Y(1) \perp\!\!\!\perp A|X$;

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(A3): (IGNORABILITY) $Y(0), Y(1) \perp\!\!\!\perp A|X$;

then

$$\mathrm{E}\, Y(1) = \mathrm{E}\left(\frac{AY}{\pi(X)}\right) \quad \text{and} \quad \mathrm{E}\, Y(0) = \mathrm{E}\left(\frac{(1 - A)Y}{1 - \pi(X)}\right)$$

Inverse Probability Weights (II)

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$$W \equiv W(X) = \frac{A}{\pi(X)} + \frac{1 - A}{1 - \pi(X)}$$

THEOREM. If $A \in \{0, 1\}$ and

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then

$$\mathrm{E}\, Y(1) = \mathrm{E}\left(\frac{AY}{\pi(X)}\right) \quad \text{and} \quad \mathrm{E}\, Y(0) = \mathrm{E}\left(\frac{(1 - A)Y}{1 - \pi(X)}\right)$$

and

$$\mathrm{ATE} = \mathrm{E}\left(\frac{AY}{\pi(X)}\right) - \mathrm{E}\left(\frac{(1 - A)Y}{1 - \pi(X)}\right). \quad (\star)$$

Inverse Probability Weights (III): Proof of the Theorem

- Suppose $A = 1$ and start from the "end".

Inverse Probability Weights (III): Proof of the Theorem

- Suppose $A = 1$ and start from the "end".
- By the consistency assumption (A1) and the law of total expectation:

$$\mathrm{E}\left(\frac{AY}{\pi(X)}\right) = \mathrm{E}\left(\frac{AY(1)}{\pi(X)}\right) = \mathrm{E}^X\,\mathrm{E}\left(\frac{AY(1)}{\pi(X)}\middle| X\right) = \mathrm{E}^X\left[\frac{1}{\pi(X)}\,\mathrm{E}\left(AY(1)|X\right)\right]$$

Inverse Probability Weights (III): Proof of the Theorem

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- By the ignorability assumption (A3) and the law of total expectation:

$$\mathrm{E}\left(\frac{AY}{\pi(X)}\right) = \mathrm{E}^X\left[\frac{1}{\pi(X)}\,\mathrm{E}\left(A|X\right)\mathrm{E}\left(Y(1)|X\right)\right]$$

Inverse Probability Weights (III): Proof of the Theorem

- Suppose $A = 1$ and start from the "end".
- By the consistency assumption (A1) and the law of total expectation:

$$\mathrm{E}\left(\frac{AY}{\pi(X)}\right) = \mathrm{E}\left(\frac{AY(1)}{\pi(X)}\right) = \mathrm{E}^X\,\mathrm{E}\left(\frac{AY(1)}{\pi(X)}\middle|X\right) = \mathrm{E}^X\left[\frac{1}{\pi(X)}\,\mathrm{E}\left(AY(1)|X\right)\right]$$

- By the ignorability assumption (A3) and the law of total expectation:

$$\mathrm{E}\left(\frac{AY}{\pi(X)}\right) = \mathrm{E}^X\left[\frac{1}{\pi(X)}\,\mathrm{E}\left(A|X\right)\mathrm{E}\left(Y(1)|X\right)\right]$$

- Since $\mathrm{E}(A|X) = \mathrm{E}(\mathbb{1}(A=1)|X) = \mathrm{P}(A=1|X) = \pi(X)$,

$$\mathrm{E}\left(\frac{AY}{\pi(X)}\right) = \mathrm{E}^X\left[\frac{1}{\pi(X)}\pi(X)\,\mathrm{E}\left(Y(1)|X\right)\right] = \mathrm{E}^X\left[\mathrm{E}\left(Y(1)|X\right)\right]$$

$$\Rightarrow \mathrm{E}\left(\frac{AY}{\pi(X)}\right) = \mathrm{E}\,Y(1).$$

Horvitz-Thompson Estimator (1)

- Formula ($\star$) suggests using the Horvitz-Thompson (HT) estimator,

$$\widehat{\mathrm{ATE}}_{\mathsf{HT}} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{A_i Y_i}{\pi(X_i)}\right) - \frac{1}{n}\sum_{i=1}^{n}\left(\frac{(1-A_i)Y_i}{1-\pi(X_i)}\right).$$

Horvitz-Thompson Estimator (2): Application (BMI Data)

NHANES: National Health and Nutrition Examination Survey (CDC)

Data: subsample from NHANES 2007–2008;

Goal: to assess the effect of the participation in school meal programs on the BMI for school children.

Variables:

```
## BMI............: (OUTCOME) Body Mass Index
## School_meal....: (TREATMENT) School meal program (1: male, 0: female)
## age............: Age
## ChildSex.......: Sex (1: male, 0: female)
## black..........: Race (1: black, 0: otherwise)
## mexam..........: Race (1: Hispanic: 0 otherwise)
## pir200_plus....: Family above 200% of the federal poverty level
## WIC............: Participation in the special supplemental nutrition program
## Food_Stamp.....: Participation in food stamp program
## fsdchbi........: Childhood food security
## AnyIns.........: Any insurance
## RefSex.........: Sex of the adult respondent (1: male, 0: female)
## RefAge.........: Age of the adult respondent
```

NHANES: Covariate Balancing Status

```
> DATA <- fread("nhanes_bmi.csv")
> vars <- c("age", "ChildSex", "black", "mexam", "pir200_plus",
+           "WIC", "Food_Stamp", "fsdchbi", "AnyIns", "RefSex", "RefAge")
> group_var <- "School_meal"
> table.obj <- tableone::CreateTableOne(vars = vars, strata = group_var, data = DATA)
> print(table.obj)

Stratified by School_meal
                            0            1          p- test
n                        1046         1284
age (mean (SD))          9.85 (4.44) 10.07 (3.51)   0.181
ChildSex (mean (SD))     0.52 (0.50)  0.51 (0.50)   0.527
black (mean (SD))        0.20 (0.40)  0.31 (0.46)  <0.001
mexam (mean (SD))        0.18 (0.38)  0.33 (0.47)  <0.001
pir200_plus (mean (SD))  0.66 (0.47)  0.25 (0.43)  <0.001
WIC (mean (SD))          0.11 (0.31)  0.26 (0.44)  <0.001
Food_Stamp (mean (SD))   0.12 (0.32)  0.44 (0.50)  <0.001
fsdchbi (mean (SD))      0.15 (0.36)  0.33 (0.47)  <0.001
AnyIns (mean (SD))       0.89 (0.32)  0.84 (0.37)   0.001
RefSex (mean (SD))       0.50 (0.50)  0.39 (0.49)  <0.001
RefAge (mean (SD))      40.29 (9.68) 38.55 (10.36) <0.001
```

$\Rightarrow$ UNBALANCED sample.

Horvitz-Thompson Estimator (4): Application (BMI Data)

APPROACH 1: Difference of Means (Naive)

```
> mean(DATA$BMI[DATA$School_meal==1]) - mean(DATA$BMI[DATA$School_meal==0])
[1] 0.5339044

> # Equivalent approach:
> summary(lm(BMI ~ School_meal, data=DATA))

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  19.8162     0.1675 118.272   <2e-16 ***
School_meal   0.5339     0.2257   2.366   0.0181 *
```

$\Rightarrow$ $\text{ATE}_{\text{NAIVE}} = 0.53 > 0.$

and $\text{IC}_{0.95}(\text{ATE}_{\text{NAIVE}}) = [0.097; 0.971].$

APPROACH 2: Linear Regression

```
> #### Linear regression approach:
> ATE_LM <- lm(BMI ~ ., data=DATA)
> summary(ATE_LM)

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) 12.7338066  0.5544644  22.966   <2e-16 ***
School_meal  0.0612479  0.2208829   0.277   0.7816
age          0.7296864  0.0256259  28.475   <2e-16 ***
ChildSex    -0.2084058  0.1886977  -1.104   0.2695
black        0.4411737  0.2390988   1.845   0.0651 .
mexam        0.6190676  0.2428658   2.549   0.0109 *
pir200_plus -0.4673003  0.2322279  -2.012   0.0443 *
WIC         -0.0007535  0.2612967  -0.003   0.9977
Food_Stamp  -0.1014021  0.2488077  -0.408   0.6836
fsdchbi     -0.1835593  0.2324985  -0.790   0.4299
AnyIns      -0.3367494  0.2833039  -1.189   0.2347
RefSex      -0.4910218  0.2002450  -2.452   0.0143 *
RefAge       0.0173300  0.0101659   1.705   0.0884 .
```

$\Rightarrow$ $\mathrm{ATE}_{\mathsf{LIN}} = 0.061 \ > 0.$

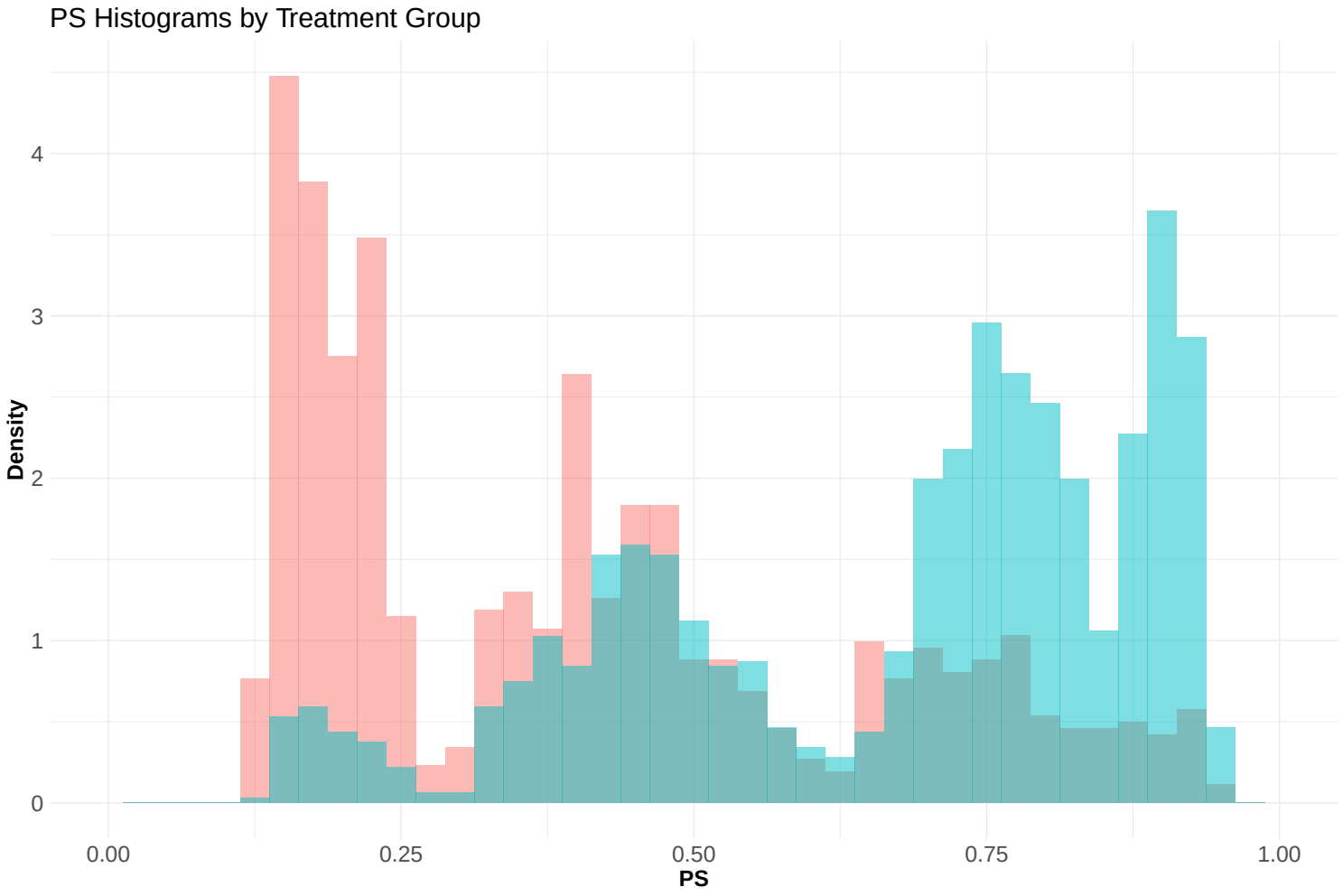
but $\mathsf{IC}_{0.95}(\mathrm{ATE}_{\mathsf{LIN}}) = [-0.372; 0.494].$

APPROACH 3: Propensity Scores

```
> #### Propensity score and IPW
> fit_PS <- glm(School_meal ~ . - BMI, data=DATA, family="binomial")
> PS <- predict(fit_PS, type="response")
> summary(PS)
Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.1338  0.3445  0.5350  0.5511  0.7794  0.9534
>
> IPW <- with(DATA, School_meal/PS + (1-School_meal)/(1-PS))
> summary(IPW)
Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1.051   1.219   1.407   2.020   2.099  21.472
>
> # Temporary data for viewing PS and IPW
> TEMP <- data.frame(A=DATA$School_meal, PS, IPW)
> head(TEMP)
  A        PS      IPW
1 0 0.1533945 1.181188
2 0 0.7924330 4.817721
3 1 0.6665961 1.500159
4 1 0.4881477 2.048560
5 0 0.7681533 4.313196
6 1 0.3384228 2.954883
```

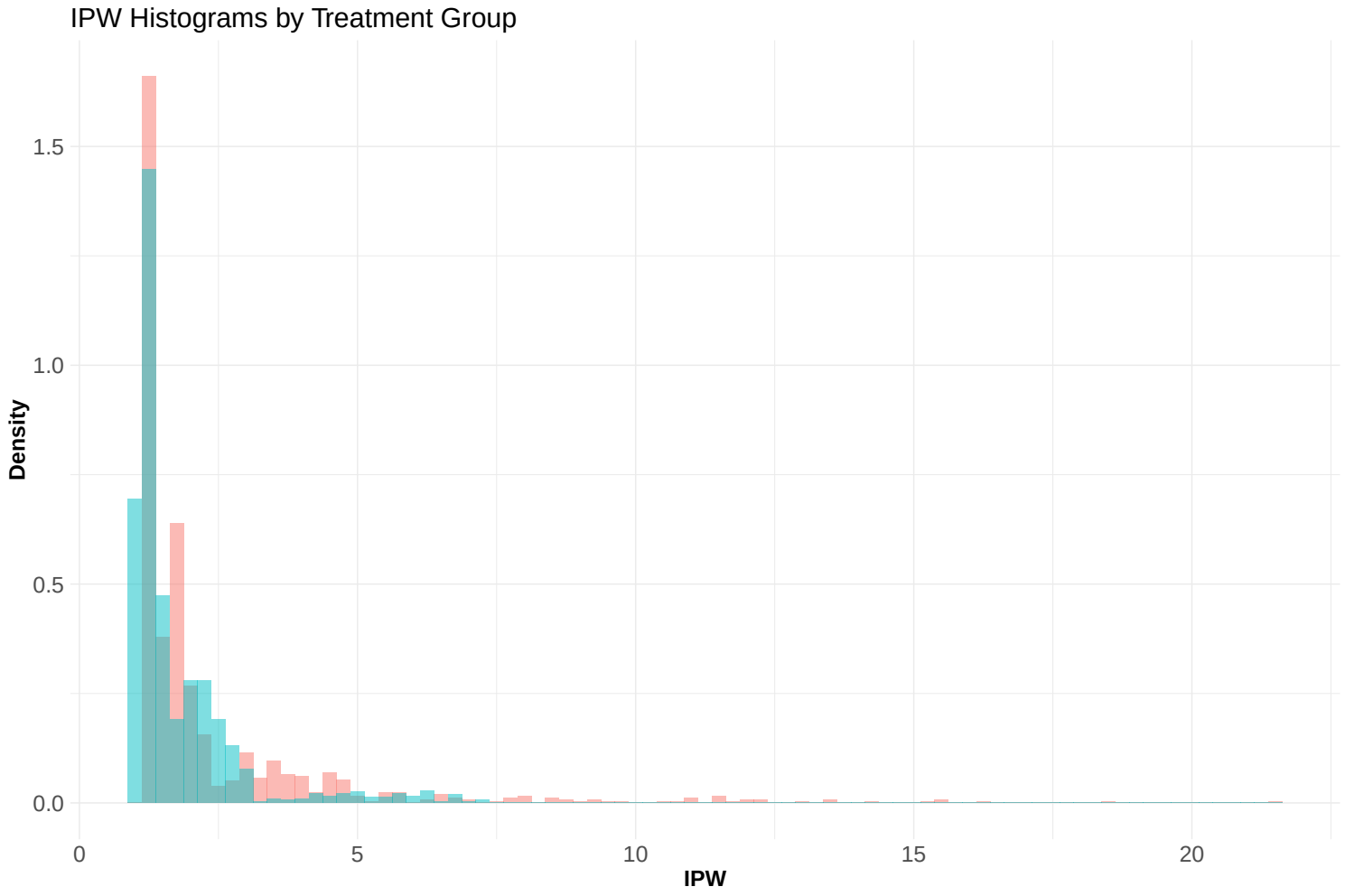
Horvitz-Thompson Estimator (7): Application (BMI Data)

APPROACH 3: Propensity Scores



Horvitz-Thompson Estimator (8): Application (BMI Data)

APPROACH 3: Propensity Scores



APPROACH 3A: Estimates Based on Propensity Scores

```
> #### Linear regression on Treatment and PS:
> ATE_LM_PS <- lm(BMI ~ School_meal + PS, data=DATA)
> summary(ATE_LM_PS)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 19.04459    0.26421  72.081  < 2e-16 ***
School_meal  0.01629    0.26364   0.062 0.950737
PS           1.91783    0.50872   3.770 0.000167 ***
```

$\Rightarrow$ $\mathrm{ATE}_{\mathsf{LINPS}} = 0.061 > 0.$

but $\mathsf{IC}_{0.95}(\mathrm{ATE}_{\mathsf{LINPS}}) = [-0.500; 0.533].$

Horvitz-Thompson Estimator (9): Application (BMI Data)

APPROACH 3A: Estimates Based on Propensity Scores

```
> #### Linear regression on Treatment and PS:
> ATE_LM_PS <- lm(BMI ~ School_meal + PS, data=DATA)
> summary(ATE_LM_PS)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 19.04459    0.26421  72.081  < 2e-16 ***
School_meal  0.01629    0.26364   0.062 0.950737
PS           1.91783    0.50872   3.770 0.000167 ***
```

$\Rightarrow$ $\mathrm{ATE}_{\mathsf{LINPS}} = 0.061 > 0.$

but $\mathsf{IC}_{0.95}(\mathrm{ATE}_{\mathsf{LINPS}}) = [-0.500; 0.533].$

```
> with(DATA, mean(School_meal*BMI*IPW) - mean((1-School_meal)*BMI*IPW))
[1] -1.516284
```

$\Rightarrow$ $\mathrm{ATE}_{\mathsf{HT}} = -1.5161 < 0.$

but $\mathsf{IC}_{0.95}(\mathrm{ATE}_{\mathsf{HT}}) = [-3.809; 0.735]$ (via bootstrap).

Day 2

Part 5: Caveats and Adjustments

Caveats

1. HT is sensitive to the positivity assumption. It will fail if:
   (a) $\pi(X_i) \approx 0$ for many treated units;
   (b) $1 - \pi(X_i) \approx 0$ for many control units.

Caveats

1. HT is sensitive to the positivity assumption. It will fail if:
   (a) $\pi(X_i) \approx 0$ for many treated units;
   (b) $1 - \pi(X_i) \approx 0$ for many control units.

2. Lack of invariance:

   Theorem. If $\widehat{\mathrm{ATE}}_{\mathsf{HT}}$ is the HT estimator for

$$\{(A_i, X_i, Y_i)\},$$

   then the HT estimator for

$$\{(A_i, X_i, Y_i + \delta)\},$$

   with δ constant, is

$$\widehat{\mathrm{ATE}}_{\mathsf{HT}} + \underbrace{\delta \left(\frac{1}{n} \sum_{i=1}^{n} \frac{A_i}{\pi(X_i)} - \frac{1}{n} \sum_{i=1}^{n} \frac{1 - A_i}{1 - \pi(X_i)} \right)}_{\text{TIPICALLY } \neq 0\ !}.$$

Adjustments and alternatives

1. Trimming reduces the sensitivity to extreme PS's:
   - DROP UNITS with $\pi(X_i) > 1 - \kappa$ or $\pi(X_i) < \kappa$ (tipically, κ is between 0.05 and 0.10).

Adjustments and alternatives

1. Trimming reduces the sensitivity to extreme PS's:
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   Truncating is another strategy to reduce the sensitivity to extreme PS's:
   - truncate the estimates for the PS's so that they are now equal to

$$\max\{\kappa,\ \min\{\pi(X_i), 1 - \kappa\}\}.$$

Adjustments and alternatives

1. Trimming reduces the sensitivity to extreme PS's:
   - DROP UNITS with $\pi(X_i) > 1 - \kappa$ or $\pi(X_i) < \kappa$ (tipically, κ is between 0.05 and 0.10).

   Truncating is another strategy to reduce the sensitivity to extreme PS's:
   - truncate the estimates for the PS's so that they are now equal to

$$\max\left\{\kappa,\ \min\left\{\pi(X_i), 1-\kappa\right\}\right\}.$$

2. Normalizing the weights removes the lack of invariance:

$$\widehat{\mathrm{ATE}}_{\text{HAJEK}} = \frac{\sum_{i=1}^{n} \frac{A_i Y_i}{\pi(X_i)}}{\sum_{j=1}^{n} \frac{A_j}{\pi(X_j)}} - \frac{\sum_{i=1}^{n} \frac{(1-A_i) Y_i}{1-\pi(X_i)}}{\sum_{j=1}^{n} \frac{1-A_j}{1-\pi(X_j)}}.$$

Example: Lalonde (2): Covariate Balancing Status

```
> vars <- c("age", "education", "black", "hispanic", "married", "nodegree",
+           "re74", "re75", "re78", "u74", "u75")
> group_var <- "treatment"
> table.obj <- tableone::CreateTableOne(vars = vars, strata = group_var,
+                                       data = lalonde)
> print(table.obj)
Stratified by treatment
                            0                  1            p-test
n                         429                185
age (mean (SD))         28.03 (10.79)      25.82 (7.16)     0.011
education (mean (SD))   10.24 (2.86)       10.35 (2.01)     0.633
black (mean (SD))        0.20 (0.40)        0.84 (0.36)    <0.001
hispanic (mean (SD))     0.14 (0.35)        0.06 (0.24)     0.003
married (mean (SD))      0.51 (0.50)        0.19 (0.39)    <0.001
nodegree (mean (SD))     0.60 (0.49)        0.71 (0.46)     0.009
re74 (mean (SD))      5619.24 (6788.75) 2095.57 (4886.62) <0.001
re75 (mean (SD))      2466.48 (3292.00) 1532.06 (3219.25)  0.001
re78 (mean (SD))      6984.17 (7294.16) 6349.14 (7867.40)  0.334
u74 (mean (SD))          0.26 (0.44)        0.71 (0.46)    <0.001
u75 (mean (SD))          0.31 (0.46)        0.60 (0.49)    <0.001
```

$\Rightarrow$ UNBALANCED sample.

Example: Lalonde (3): Already Known Estimates

Estimate 1: Difference of Means (on the Original/Randomized Data)

$$\text{ATE}_{\text{RAND}} = 1,794.34, \quad \text{C.I}(95\%) = [994.66;\ 2.594.02].$$

Estimate 2: Difference of Means (Naive)

$$\text{ATE}_{\text{NAIVE}} = -635.03, \quad \text{C.I}(95\%) = [-1,967.24;\ 697.19].$$

Estimate 3: Regression only on `treat` and `age`

$$\text{ATE}_{\text{REG_PARTIAL}} = -462.060, \quad \text{C.I}(95\%) = [-1,751.107;\ 193.657].$$

Estimate 4: Regression only on all covariates

$$\text{ATE}_{\text{REG_FULL}} = 1,596.320, \quad \text{C.I}(95\%) = [68.232;\ 3,124.637].$$

Example: Lalonde (4): Propensity Scores

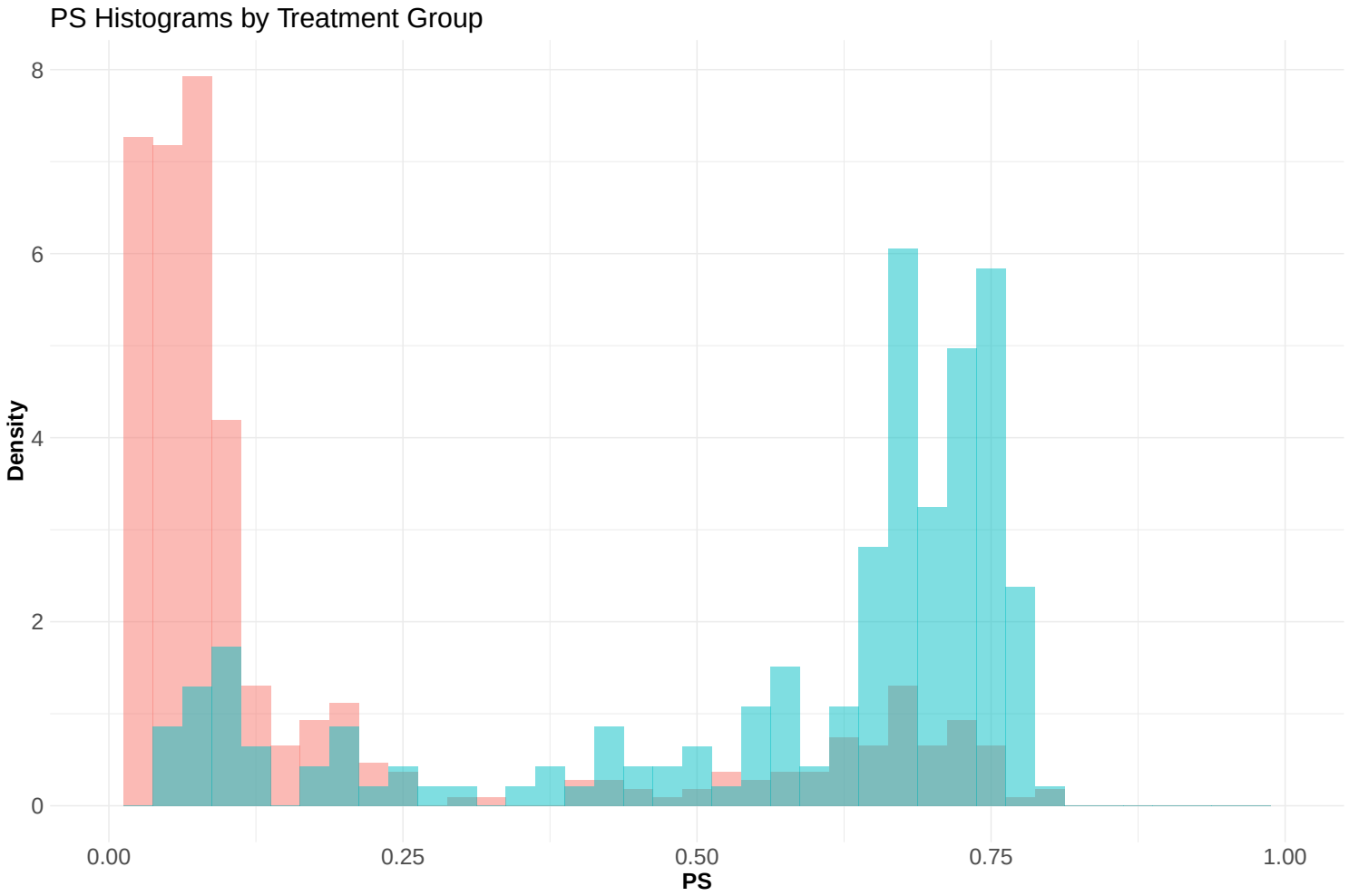
APPROACH 3:

```
> PS_glm <- glm(treatment ~ age + education + I(education^2) + I(education^3) +
+                factor(nodegree) + factor(black) + factor(hispanic) + re74 + re75,
+               data=lalonde, family="binomial")
> summary(PS_glm)

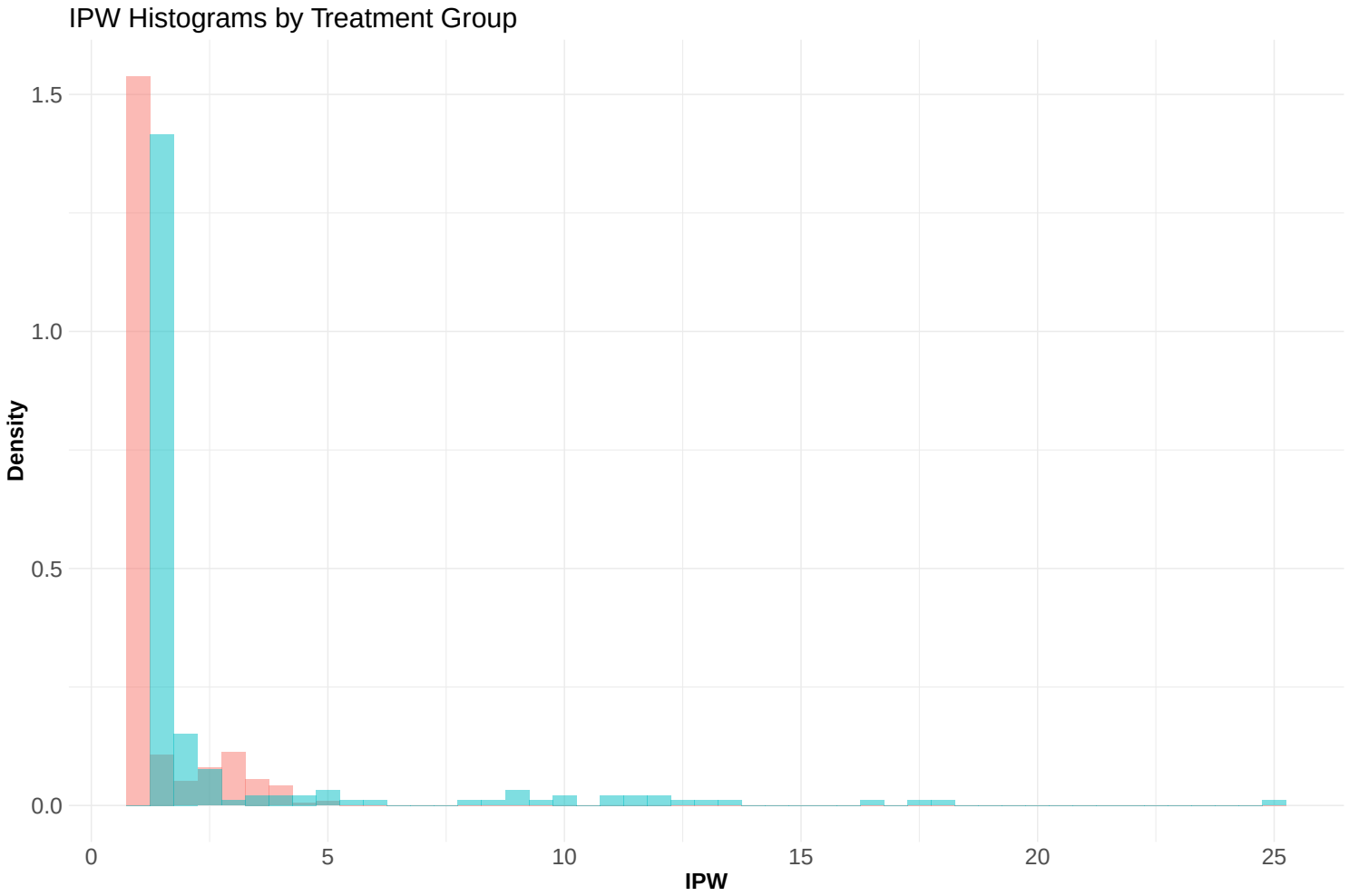
Coefficients:
                    Estimate Std. Error z value Pr(>|z|)
(Intercept)       -5.892e+00  2.167e+00  -2.719  0.00654 **
age                1.009e-02  1.334e-02   0.756  0.44955
education          3.351e-01  7.431e-01   0.451  0.65205
I(education^2)     1.300e-02  8.218e-02   0.158  0.87429
I(education^3)    -1.547e-03  2.835e-03  -0.546  0.58540
factor(nodegree)1  3.897e-01  4.005e-01   0.973  0.33055
factor(black)1     3.167e+00  2.842e-01  11.145  < 2e-16 ***
factor(hispanic)1  1.029e+00  4.192e-01   2.453  0.01415 *
re74              -7.771e-05  2.861e-05  -2.716  0.00660 **
re75               1.910e-05  4.452e-05   0.429  0.66792

> PS <- as.numeric(predict(PS_glm, type="response"))
>
> IPW <- as.numeric(lalonde$treat/PS + (1-lalonde$treat)/(1-PS))
```

Example: Lalonde (5): Propensity Scores



Example: Lalonde (6): IPW's



Example: Lalonde (7) — Estimates Based on PS and IPW

Estimate 5: Horvitz-Thompson

```
> with(lalonde, mean(treatment*re78*IPW) - mean((1-treatment)*re78*IPW))
[1] -337.5471
```

$$\text{ATE}_{\text{HT}} = -337.547, \quad \text{C.I}(95\%) = [-1,969.0;\ 1,744.9] \quad \text{(BOOTSTRAP)}.$$

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Estimate 6: Hajek

```
> with(lalonde, (sum(treatment*re78*IPW)/sum(treatment*IPW)) -
+         (sum((1-treatment)*re78*IPW))/sum((1-treatment)*IPW))
[1] 481.6272
```

$$\text{ATE}_{\text{HJ}} = 481.627, \quad \text{C.I}(95\%) = [-1,337.3;\ 2,637.2] \quad \text{(BOOTSTRAP)}.$$

Example: Lalonde (8) — Estimates Based on PS and IPW

Estimate 6: Trimming (Horvitz-Thompson)

```
> Q <- 0.05
> lalonde_trim <- lalonde[PS>Q & PS<(1-Q),]
> PS_trim <- PS[PS>Q & PS<(1-Q)]
> IPW_trim <- as.numeric(lalonde_trim$treat/PS_trim +
+                          (1-lalonde_trim$treat)/(1-PS_trim))
> with(lalonde_trim, mean(treat*re78*IPW_trim) - mean((1-treat)*re78*IPW_trim))
[1] 2280.127
```

$$\text{ATE}_{\text{TRIM}} = 2,280.127, \quad \text{C.I}(95\%) = [71;\ 4,069] \quad \text{(BOOTSTRAP)}.$$

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[1] 2280.127
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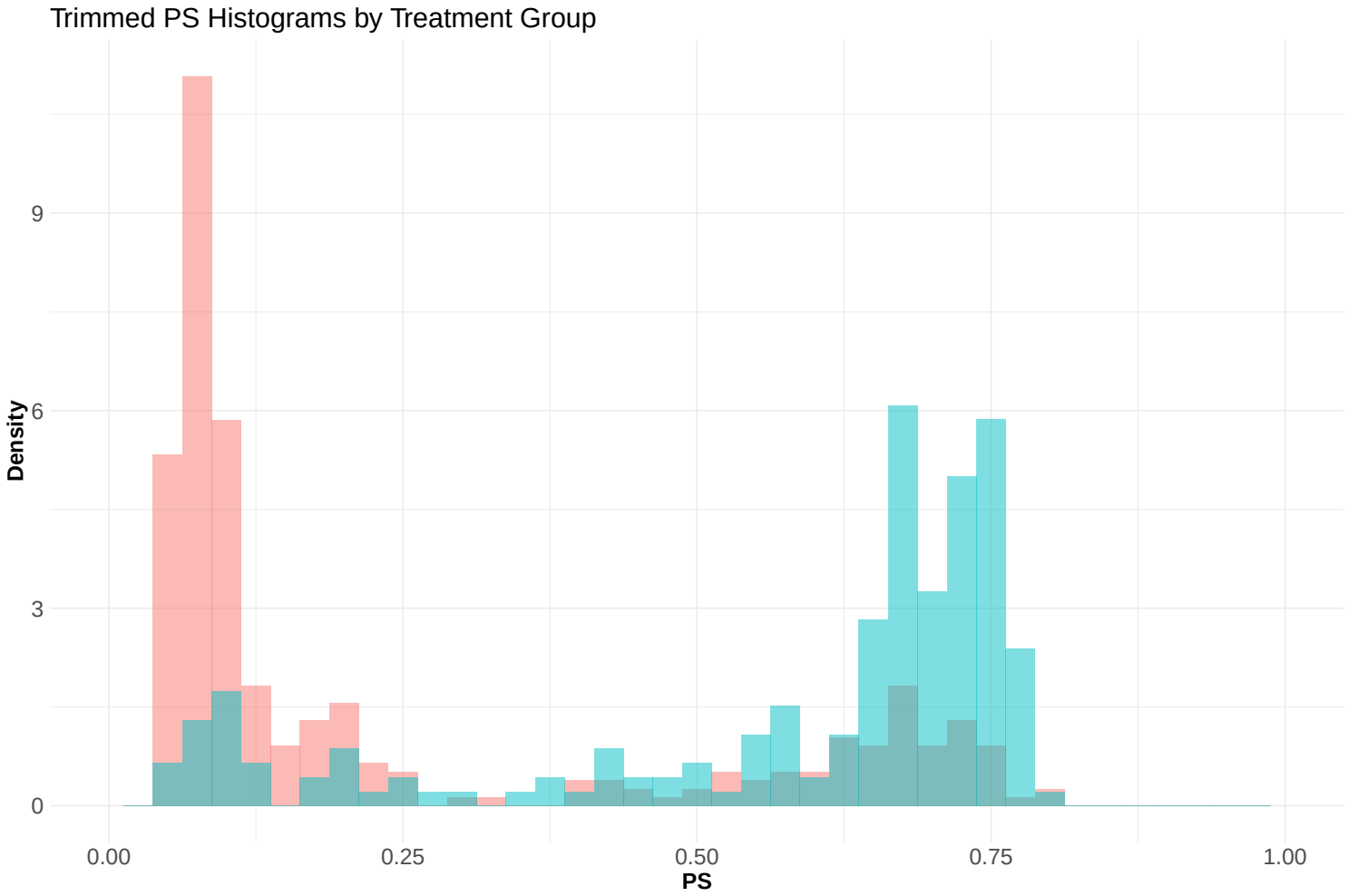
Estimate 6: Trimming (Hajek)

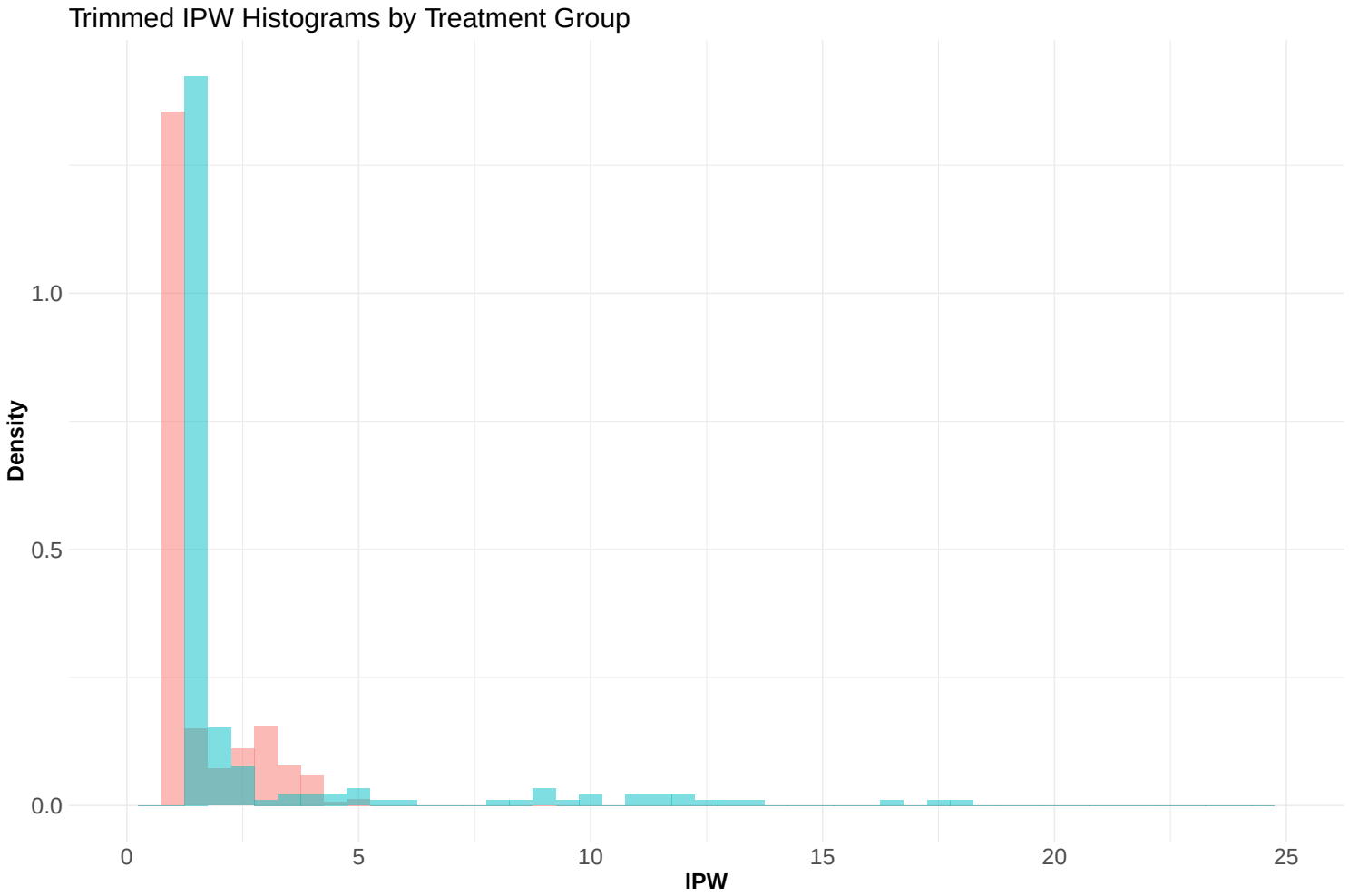
```
> with(lalonde_trim, (sum(treatment*re78*IPW_trim)/sum(treatment*IPW_trim)) -
+        (sum((1-treatment)*re78*IPW_trim))/sum((1-treatment)*IPW_trim))
[1] 1880.384
```

$$\text{ATE}_{\text{TRIM}} = 1,880.384, \quad \text{C.I}(95\%) = [-34;\ 3,393] \quad \text{(BOOTSTRAP)}.$$

Example: Lalonde (9): Trimmed Propensity Scores



Example: Lalonde (10): Trimmed IPW's



Day 2

Part 6: MSM and DR Estimators

Other Estimators (1A): Marginal Structural Models (MSM)

- Suppose

$$\mathrm{E}\, Y(a) = \beta_0 + \beta_1 a.$$

$\Rightarrow$ $\mathrm{ATE} = \mathrm{E}\, Y(1) - \mathrm{E}\, Y(0) = \beta_1 \quad (\star).$

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- Models for the marginal mean of a counterfactual outcome are referred to as MSM.
- Remark 1: alternatively, if the outcome is binary, we could consider

$$\mathrm{logit}\, \mathrm{E}\, Y(a) = \beta_0 + \beta_1 a.$$

- Remark 2: the formulation $(\star)$ is not the only one. For example, we could assume

$$\mathrm{E}\, Y(a) = \beta_0 + \beta_1 a + \beta_2 a^2.$$

(If we believe that the dose-response curve is better represented by a parabola)

- Remark 3: EFFECT MODIFICATION

$$\mathrm{E}[Y(a)|S] = \beta_0 + \beta_1 a + \beta_2 aS + \beta_3 S.$$

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$$\mathrm{E}[Y|A] = \alpha_0 + \alpha_1 A,$$

let

$$(\widehat{\beta}_0, \widehat{\beta}_1) = \underset{\alpha_0, \alpha_1}{\mathrm{argmin}} \sum_{i=1}^{n} \widetilde{W}_i \left(Y_i - \alpha_0 - \alpha_1 A_i\right)^2,$$

where

$$\widetilde{W}_i = \frac{\mathrm{E}\,A}{\pi(X_i)} \quad \text{(STABILIZED WEIGTHS)}$$

Other Estimators (1B): Marginal Structural Models (MSM)

- Remark 3: EFFECT MODIFICATION

$$\mathrm{E}[Y(a)|S] = \beta_0 + \beta_1 a + \beta_2 aS + \beta_3 S.$$

In this case,

$$\mathrm{ATE}(s) = \mathrm{E}[Y(1)|S = s] - \mathrm{E}[Y(0)|S = s] = \beta_1 + \beta_2 s.$$

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$$\mathrm{E}[Y|A] = \alpha_0 + \alpha_1 A,$$

let

$$(\widehat{\beta}_0, \widehat{\beta}_1) = \underset{\alpha_0, \alpha_1}{\mathrm{argmin}} \sum_{i=1}^{n} \widetilde{W}_i \left(Y_i - \alpha_0 - \alpha_1 A_i\right)^2,$$

where

$$\widetilde{W}_i = \frac{\mathrm{E}\,A}{\pi(X_i)} \quad \text{(STABILIZED WEIGTHS)}$$

- Remark 4: in the case of effect modification, the weights are

$$\widetilde{W}_i = \frac{\mathrm{E}[A|S]}{\pi(X_i)} \quad \text{(HENCE, } S \text{ SHOULD NOT BE EQUAL TO } X\text{)}$$

Other Estimators (2A): Double Robust Estimators (DR)

- Let

$$\mu_a(X;\beta_a) = \mathrm{E}[Y|A=a,X;\beta_a]$$

and define

$$m_1 = \mathrm{E}\left[\frac{AY}{\pi(X;\theta)} - \frac{A-\pi(X;\theta)}{\pi(X;\theta)}\mu_1(X;\beta_1)\right].$$

$$m_0 = \mathrm{E}\left[\frac{(1-A)Y}{1-\pi(X;\theta)} - \frac{\pi(X;\theta)-A}{1-\pi(X;\theta)}\mu_0(X;\beta_0)\right].$$

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$$m_0 = \mathrm{E}\left[\frac{(1-A)Y}{1-\pi(X;\theta)} - \frac{\pi(X;\theta)-A}{1-\pi(X;\theta)}\mu_0(X;\beta_0)\right].$$

or, equivalently,

$$m_1 = \mathrm{E}\left[\frac{A(Y-\mu_1(X;\beta_1))}{\pi(X;\theta)} + \mu_1(X;\beta_1)\right].$$
$$m_0 = \mathrm{E}\left[\frac{(1-A)(Y-\mu_1(X;\beta_0))}{1-\pi(X;\theta)} + \mu_0(X;\beta_0)\right].$$

Other Estimators (2B): Double Robust Estimators (DR)

THEOREM: If

- either $\pi(X;\theta)$ or $(\mu_0(X;\beta_0), \mu_1(X;\beta_1))$ is correctly specified,

then

$$\mathrm{ATE} = m_1 - m_0.$$

Other Estimators (2B): Double Robust Estimators (DR)

THEOREM: If

- either $\pi(X;\theta)$ or $(\mu_0(X;\beta_0), \mu_1(X;\beta_1))$ is correctly specified,

then

$$\mathrm{ATE} = m_1 - m_0.$$

In general, if

- either $\pi(X;\theta)$ or $\mu_a(X;\beta_a)$ is correctly specified,

then

$$\mathrm{E}\, Y(a) = m_a.$$

Other Estimators (2C): Double Robust Estimators (DR)

PROOF:

$$
\begin{aligned}
m_1 - \mathrm{E}\,Y(1) &= \mathrm{E}\left[\frac{A(Y-\mu_1(X;\beta_1))}{\pi(X;\theta)} + \mu_1(X;\beta_1) - Y(1)\right] \\
&= \mathrm{E}\left[\frac{A(Y(1)-\mu_1(X;\beta_1))}{\pi(X;\theta)} - (Y(1)-\mu_1(X;\beta_1))\right] \qquad \text{(CONSISTENCY)} \\
&= \mathrm{E}\left[\frac{A-\pi(X;\theta)}{\pi(X;\theta)}(Y(1)-\mu_1(X;\beta_1))\right] \\
&= \mathrm{E}^X\,\mathrm{E}\left[\frac{A-\pi(X;\theta)}{\pi(X;\theta)}(Y(1)-\mu_1(X;\beta_1))\middle| X\right] \\
&= \mathrm{E}^X\left(\mathrm{E}\left[\frac{A-\pi(X;\theta)}{\pi(X;\theta)}\middle| X\right]\mathrm{E}\left[Y(1)-\mu_1(X;\beta_1)|X\right]\right) \qquad \text{(IGNORABILITY)} \\
&= \mathrm{E}^X\left(\frac{\pi(X)-\pi(X;\theta)}{\pi(X;\theta)}(\mu_1(X)-\mu_1(X;\beta_1))\right) \\
&= 0 \quad \Leftrightarrow \quad \pi(X;\theta)=\pi(X) \quad \text{or} \quad \mu_1(X;\beta_1)=\mu_1(X).
\end{aligned}
$$

Day 2

Part 7: ATT

Average Treatment Effect on Treated (ATT)

$$\mathrm{ATT} = \mathrm{E}[Y(1) - Y(0)|A = 1].$$

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Identification of ATT via g-Formula:

$$\begin{aligned}
\mathrm{ATT} &\overset{\text{Def}}{=} \mathrm{E}[Y(1)|A = 1] - \mathrm{E}[Y(0)|A = 1] \\
&= \mathrm{E}[Y|A = 1] - \mathrm{E}^{X|A=1}\,\mathrm{E}[Y(0)|A = 1, X] \\
&\overset{\text{Ign}}{=} \mathrm{E}[Y|A = 1] - \mathrm{E}^{X|A=1}\,\mathrm{E}[Y(0)|A = 0, X] \\
&\overset{\text{Con}}{=} \mathrm{E}[Y|A = 1] - \mathrm{E}^{X|A=1}\,\mathrm{E}[Y|A = 0, X]
\end{aligned}$$

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\end{aligned}$$

$$\therefore \text{ATE} = \text{ATT} \quad \Leftrightarrow \quad \text{E}^{X|A=1}\,\text{E}[Y|A = 1, X] = \text{E}[Y|A = 1].$$

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\end{aligned}$$

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g-FORMULA estimator:

$$\widehat{\mathrm{ATT}} = \frac{1}{n_1} \sum_{i:A_i=1} Y_i - \frac{1}{n_1} \sum_{i:A_i=1} \widehat{\mathrm{E}}[Y|A = 0, X_i]$$

IPW for ATT

FACT:

$$\mathrm{ATT} = \frac{1}{\mathrm{E}\,A}\left(\mathrm{E}\left[AY\right] - \mathrm{E}\left[\frac{\pi(X)}{1-\pi(X)}(1-A)Y\right]\right) = \frac{1}{\mathrm{E}\,A}\,\mathrm{E}\left[\frac{A-\pi(X)}{1-\pi(X)}Y\right].$$

IPW for ATT

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"IPW" ESTIMATOR:

- Taking

$$\widehat{\mathrm{E}}A = \widehat{\mathrm{P}}(A=1) = \frac{n_1}{n},$$

$$\Rightarrow \widehat{\mathrm{ATT}} = \frac{1}{n}\sum_{i=1}^{n} A_i Y_i - \frac{1}{n}\sum_{i=1}^{n}\frac{\pi(X_i)}{1-\pi(X_i)}(1-A_i)Y_i$$

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- Remark 1: the IPW's for ATT are

$$W_i = \mathbb{1}(A_i = 1) + \mathbb{1}(A_i = 0)\frac{\pi(X_i)}{1-\pi(X_i)}$$

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- Remark 1: the IPW's for ATT are

$$W_i = \mathbb{1}(A_i = 1) + \mathbb{1}(A_i = 0)\frac{\pi(X_i)}{1-\pi(X_i)}$$

- Remark 2: the positivity condition for ATT is just $1 - \pi(X) > 0$.

ATE vs. ATT

$\underline{\text{ATE} \neq \text{ATT}}$:

ATE vs. ATT

$\text{ATE} \neq \text{ATT}$:

- Example:

$$Y = \beta_0 + \tau A + \beta_1 X + \varepsilon \quad \Rightarrow \quad \text{ATE} = \tau.$$

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- Example:

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On the other hand, $\mathrm{E}[Y|A = 0, X] = \beta_0 + \beta_1 X$

$$\Rightarrow \mathrm{E}^X \, \mathrm{E}[Y|A = 0, X] = \beta_0 + \beta_1 \, \mathrm{E}\, X.$$

ATE vs. ATT

$\text{ATE} \neq \text{ATT}$:

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$$Y = \beta_0 + \tau A + \beta_1 X + \varepsilon \quad \Rightarrow \quad \text{ATE} = \tau.$$

On the other hand, $\mathrm{E}[Y|A = 0, X] = \beta_0 + \beta_1 X$

$$\Rightarrow \mathrm{E}^X \, \mathrm{E}[Y|A = 0, X] = \beta_0 + \beta_1 \, \mathrm{E}\, X.$$

and $\mathrm{E}[Y|A = 1] = \beta_0 + \tau + \beta_1 \, \mathrm{E}[X|A = 1]$

$$\begin{aligned} \Rightarrow \text{ATT} &= (\beta_0 + \tau + \beta_1 \, \mathrm{E}[X|A = 1]) - (\beta_0 + \beta_1 \, \mathrm{E}\, X) \\ &= \tau + \beta_1 \, \{\mathrm{E}[X|A = 1] - \mathrm{E}\, X\} \\ &= \text{ATE} + \beta_1 \{\mathrm{E}[X|A = 1] - \mathrm{E}\, X\} \end{aligned}$$

ATE vs. ATT

$\text{ATE} \neq \text{ATT}$:

- Example:

$$Y = \beta_0 + \tau A + \beta_1 X + \varepsilon \quad \Rightarrow \quad \text{ATE} = \tau.$$

On the other hand, $\mathrm{E}[Y|A = 0, X] = \beta_0 + \beta_1 X$

$$\Rightarrow \mathrm{E}^X \, \mathrm{E}[Y|A = 0, X] = \beta_0 + \beta_1 \, \mathrm{E}\, X.$$

and $\mathrm{E}[Y|A = 1] = \beta_0 + \tau + \beta_1 \, \mathrm{E}[X|A = 1]$

$$\begin{aligned} \Rightarrow \text{ATT} &= (\beta_0 + \tau + \beta_1 \, \mathrm{E}[X|A = 1]) - (\beta_0 + \beta_1 \, \mathrm{E}\, X) \\ &= \tau + \beta_1 \, \{\mathrm{E}[X|A = 1] - \mathrm{E}\, X\} \\ &= \text{ATE} + \beta_1 \{\mathrm{E}[X|A = 1] - \mathrm{E}\, X\} \end{aligned}$$

Therefore,

$$\text{ATT} = \text{ATE} \quad \Leftrightarrow \quad \mathrm{E}[X|A = 1] = \mathrm{E}\, X$$

ATE vs. ATT

Given $X = x$, then $\mathrm{ATE}(x) = \mathrm{ATT}(x)$:

- Os efeitos condicionais $\mathrm{ATT}(x) = \mathrm{ATE}(x)$.

$$
\begin{aligned}
\mathrm{ATT}(x) &= \mathrm{E}[Y(1)|A = 1, X = x] - \mathrm{E}[Y(0)|A = 1, X = x] \\
&= \mathrm{E}[Y|A = 1, X = x] - \mathrm{E}[Y(0)|A = 0, X = x] \\
&= \mathrm{E}[Y|A = 1, X = x] - \mathrm{E}[Y|A = 0, X = x] \\
&= \mathrm{ATE}(x).
\end{aligned}
$$

ATE vs. ATT

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&= \mathrm{E}[Y|A = 1, X = x] - \mathrm{E}[Y|A = 0, X = x] \\
&= \mathrm{ATE}(x).
\end{aligned}$$

In randomized studies $\mathrm{ATE} = \mathrm{ATT}$:

- Randomization implies $(Y(0), Y(1) \perp\!\!\!\perp A)$

$$\Rightarrow \mathrm{ATT} = \mathrm{E}[Y(1)|A = 1] - \mathrm{E}[Y(0)|A = 1] = \mathrm{E}[Y(1)] - \mathrm{E}[Y(0)] = \mathrm{ATE}$$

Example: Lalonde (1)

RECALLING THE DATA:

```
> data(lalonde, package = "Matching")
> head(lalonde)

age educ black hisp married nodegr re74 re75     re78 u74 u75 treat
1  37   11     1    0       1      1    0    0  9930.05   1   1     1
2  22    9     0    1       0      1    0    0  3595.89   1   1     1
```

QUESTION: What is the causal effect of the treatment on the treated?

- i.e., the effect of the job training program on the units who received this training?

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QUESTION: What is the causal effect of the treatment on the treated?

- i.e., the effect of the job training program on the units who received this training?

PROXY: Difference of means (using randomized data)

```
> summary(lm(re78 ~ treat, data=lalonde_rand))

Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)   4554.8      408.0  11.162  < 2e-16 ***
treat         1794.3      632.9   2.835  0.00479 **
```

$$\text{ATT}_{\text{RAND}} = 1,794.3, \quad \text{C.I}(95\%) = [553.8;\ 3,034.8].$$

Example: Lalonde (2)

Estimate 1: Based on g-formula and regresssion

```
lalonde_fit <- lm(re78 ~ treatment + age +
                     education + I(education^2) + I(education^3) +
                     factor(nodegree) + factor(black) + factor(hispanic) +
                     re74 + re75, data=lalonde)

TEMP <- lalonde
TEMP$treatment <- 0
lalonde_pred <- predict(lalonde_fit, newdata=TEMP)
```

$$\text{ATT}_{\text{RAND}} = 1,596.3, \quad \text{C.I}(95\%) = [187;\ 3,046] \quad \text{(BOOTSTRAP)}.$$

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```
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```

$$\text{ATT}_{\text{RAND}} = 1,596.3, \quad \text{C.I}(95\%) = [187;\ 3,046] \quad \text{(BOOTSTRAP)}.$$

Estimate 2: IPW

```
with(lalonde_trim,
     (sum(treatment*re78*IPW_trim)/sum(treatment*IPW_trim)) -
     (sum((1-treatment)*re78*IPW_trim))/sum((1-treatment)*IPW_trim))
```

$$\text{ATT}_{\text{IPW}} = 1,493.52, \quad \text{C.I}(95\%) = [-145;\ 3,172] \quad \text{(BOOTSTRAP)}.$$

Thank you!